ALGORITHMIC GEOPOLITICS: METHODOLOGY OF AI-DRIVEN REAL-TIME STABILITY INDEXING WITHIN THE NATIONFILES FRAMEWORK

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ABSTRACT

Background: Geopolitical information systems must reconcile high signal diversity from open and semi-open source contexts (**OSINT**) with traceable aggregation, without allowing semantic drift between raw signal, analytical index, and public presentation.

Subject: We describe the **NationFiles** framework as a hybrid situational awareness and comparison platform: an operationally advancing data pipeline, a multi-stage documented **NationFiles Stability Index (NFSI)**, and a pluralised **controller surface** that projects the same relational truth into multiple **presentation ontologies**. We further situate the **Naciro Intelligence Engine** and the **LPU-oriented inference architecture** documented in the public **Knowledge Graph** (Large Processing Unit — not vendor-specific hardware as the definition) within the overall architecture.

Methods: At the core lies a **three-stage stability pipeline** (normalisation, day-level aggregation, weighted end composition) with explicit treatment of missing values, domain logics, and rule-based coupling — the NFSI is framed as a **descriptive, rule-based aggregate**, not as an autonomous prognostic "judgment."

Integrity: The **integrity strategy** stresses, inter alia, avoiding empty navigation promises, cartographic restraint toward pseudo-precision, synchronisation of structured metadata with knowledge definitions, and **transparency over elegance**.

Keywords: Geopolitics; OSINT; stability index; data pipeline; knowledge graph; governance; open science; persistent identifier; citation practice

Strategic core positions (for audit and peer review)

1. **The NFSI is not an oracle.** It is consistently framed as a **descriptive, rule-based aggregate**; any forecast or action claims — if delivered at all — must be **explicitly** separated from the index logic and versioned.
2. **Transparency around data gaps.** Stage 2 uses documented **recovery rules** so that missing or thin inputs are **not** silently misread as low-risk or "peaceful" normality.

3. **Integrity strategy.** The operational principle **"transparency over elegance"** prioritises **honest ambiguity** and visible assumptions over smooth but misleading surfaces — central for scientific and regulatory audit paths.

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1. INTRODUCTION AND PARADIGM SHIFT

Classical geopolitical situational pictures were long produced mainly in an **archiving**, **delayed** mode. National and international decision processes today face **higher temporal expectations** — alongside growing complexity from heterogeneous data sources. The framework described here therefore follows an **operational** paradigm: raw signals are continuously ingested, cleaned, normalised, and converted into **evaluation and presentation aggregates**; the public surface reflects the same headline metrics in maps, profiles, tables, and exports without permitting silent semantic drift (cf. NationFiles Research, 2026, Part A.1) [^monographie].

The platform functionally combines **characteristics** often seen separately in public perception: statistical structure (macro-economic and governance-related modules), encyclopaedically curated context (knowledge graph), and **high-frequency** updating of situational and security views — always under the constraint of **explainable provenance** via registers, layer texts, and status reports.

1.1 Problem statement: Semantic drift as a systemic risk

For reviewers in agencies and academia, **semantic drift** — gradual divergence between raw signal, internal evaluation logic, and publicly visible headline figures — is harder to prove than a single arithmetic error. NationFiles addresses drift through **a single canon**: the same relational truth is projected into multiple **presentation ontologies**, not multiply recomputed **independently** (NationFiles Research, 2026, Parts F, J) [^monographie]. This design is **citation-friendly**: a citation of the NFSI remains compatible with a citation of layer documentation as long as version discipline holds.

1.2 Distinction from purely static information offerings

Pure encyclopaedias explain **terms**, not necessarily **situations**. Pure news aggregates narrate **events**, not necessarily **comparable** country states over time. The framework combines **term and situation logic** without one replacing the other: the knowledge graph fixes definitions; the NFSI materialises daily aggregated situation; controllers choose the **public-facing** interface per audience (Parts C–F) [^monographie].

1.3 Contribution of this whitepaper

This manuscript **distils** the internal monograph into an argument **suited to persistent identifiers**. It does not replace full architecture documentation; it grounds the **methods and governance baseline** external citation should refer to — especially stages 1–3 (Part B.2), ontology inventory (Part J), and the integrity strategy (Part W) [^monographie].

[Figure 1: Paradigm shift — from static reading stock to continuously advanced situational picture; role of connectors, pipeline, and presentation layers]

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2. ARCHITECTURE AND INFRASTRUCTURE

2.1 Backend: Connector ecosystem and operational discipline

On the input side stand hundreds of specialised **connectors**, organised as specialisations of a common execution model. Each connector has defined fetch intervals, lock logic, and target artefacts in relational materialisation. A scheduler caps total runtime and parallelism; optionally a **FIFO job queue** supports strict ordering and diagnostics for **stuck** jobs (NationFiles Research, 2026, Part B.1) [^monographie].

This layer is the **epistemic foundation**: without a disciplined connector ecosystem there is no explainable NFSI — only a pile of loose tables.

2.2 Naciro Intelligence Engine and Knowledge Graph

In the public **Knowledge Graph** (HTTPS-based entity planes), **Naciro** as analytical system engine and **NFSI** as central stability indicator are definitionally anchored; terms such as **Engine**, **LPU** (Large Processing Unit — in the graph an **architecture entity**, not a marketing label), and **Core Hierarchy** are semantically supported so that citation and internal pipeline share the same conceptual basis (cf. knowledge entities; formal treatment in the companion NationFiles/Naciro methodology publication) [^methodik].

Naciro is described there as the engine executing the documented renewal cycle of the platform and NFSI-conformant evaluation transformations; upstream lie published connectors and profiles, downstream materialised fields for maps and dashboards [^methodik]. For **LPU**, the graph documents a **specialised inference architecture** with **low latency** (the companion text cites sub-50 ms inference as a published order of magnitude) and **deterministic embedding** relative to the overall architecture — without vendor-specific accelerator hardware as the definitional core [^methodik].

2.3 Frontend: Multi-controller orchestration

The visible web application is not a monolithic blog but an **orchestration stack** of modular **controllers** serving URL spaces, translations, export channels, and visualisation families. A **base controller** supplies global state (multilingualism, canonicalisation of territorial codes, world stability mappings, consistent colour logic for vector country maps) before domain controllers load their modules (NationFiles Research, 2026, Parts A.3, C.1) [^monographie].

2.4 Geometry, auxiliary systems, and operational sustainability

Beyond the core pipeline, **auxiliary systems** exist for vector geometry (web maps), imagery (globe illustrations), migration data families, and **maintenance cycles** (health checks, cleanup of

temporary import artefacts). This layer matters for the whitepaper because **cartographic integrity** and **performance** are part of the epistemic claim: large choropleth maps are not "neutrally pretty" but semantically loaded if they hide data gaps (NationFiles Research, 2026, Parts B.3, D.1) [^monographie].

[Figure 2: Architecture overview — connectors → materialisation → engine/pipeline → controller ontologies → export & structured data]

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3. METHODS: THE THREE-STAGE STABILITY PIPELINE (CORE)

3.1 Logical role of the three stages

The pipeline is the **mathematical–logical heart** of the indexing (Part B.2 of the monograph) [^monographie]. Here the stages are described **functionally**, not implementation-specifically:

Stage 1 — Raw signal normalisation: Incoming raw rows are transformed **per source** to a unified 0–100 scale. **Directional sensitivity** ("higher is worse"), linear impact factors, and **selective updates** are applied so noise does not destabilise every state.

Stage 2 — Day aggregation at country level: Per country and day, normalised contributions are aggregated; **today's and yesterday's** day values are combined with **documented weighting**. Rules address **recovery** for missing days and **conservative starting values** for security domains — an epistemically central point: absence of data must not silently be read as "peaceful."

Stage 3 — Weighted end composition to country score: Connector contributions are weighted; missing contributions are **neutrally substituted** (in the sense of defined substitution discipline, not political neutrality rhetoric). Further documented rule families include conflict and fragility logics, population-related add-on rules, institutional coupling, and caps. Non-country or static connectors are excluded; currency nodes are handled as **virtual** nodes linked to country assignment.

3.2 NFSI as a descriptive, rule-based aggregate

The **NFSI** is thus a **rule-based, traceably documented aggregate** over heterogeneous inputs — not a singular ML "score" in the black-box prognosis sense. Prognostic or exploratory text elements on the product surface must be **explicitly** labelled and remain separably readable from index logic (integrity strategy; NationFiles Research, 2026, Part W) [^monographie].

3.3 OSINT signals and source heterogeneity

OSINT-style strands (media, event corpora, open registers) enter as **connector families** under the same normalisation and gatekeeping rules as more structured macro series. The public security/radar layer uses **filtering** instances (gatekeepers) to reduce duplicates and echo bias —

with formally documented responsibility regarding **false-negative** and human-rights sensitivity (NationFiles Research, 2026, Parts C.3, C.7) [^monographie].

3.4 Provisional algebra of Stage 3 (pseudocode; implementation-agnostic)

The purpose of this section is to give Stage **3** an **auditable** structure without anticipating proprietary code paths. Symbols and function names are **methodological placeholders**; concrete parameter values are taken from the monograph's **weighting and layer register** or from released configuration artefacts (NationFiles Research, 2026, Part B.2) [^monographie].

Notation (per country (c), calendar day (t), connector (k) from the set of country-eligible, non-static connectors (\mathcal{K}_{c,t})):

SYMBOL	MEANING
(x_{c,k,t}\in[0,100])	**Normalised** day contribution of connector (k) after stages 1–2
(\delta_{c,k,t}\in{0,1})	**Availability** (1 = record present and valid)
(w_k \ge 0)	**Base weight** from documented register (after normalisation (\sum_{k\in\mathcal{K}} w_k' = 1) for the active subset)
(\eta \in [0,100])	**Substitution level** for missing contributions — *not* "political neutrality" but a **defined** replacement metric in the rulebook
(\pi_{c,k,t} \in (0,1])	**Population scaling** (family-specific documented function of demography/exposure)
(\iota_{c,k,t} \in (0,1])	**Institutional coupling** (attachment to documented governance signals; 1 = no up/down adjustment)
(\mu^{\mathrm{con}}*{c,t}, \mu^{\mathrm{frag}}*{c,t}\in [1,\mu_{\max}])	**Conflict** and **fragility** multipliers ("malus" families)
(U_c)	**Upper bound** on country score after all terms (documentation of "cap" semantics)

Weight vector before malus (per active connector):

[\omega_{c,k,t} \;=\; w*k' \cdot \pi*{c,k,t} \cdot \iota_{c,k,t}\,.]

Effective input under substitution:

[\tilde{x}*{c,k,t} \;=\; \delta*{c,k,t}\, x*{c,k,t} \;+\; (1-\delta*{c,k,t})\, \eta\,.]

Malus application (sketch as composable mapping): The monograph treats **conflict** and **fragility** logics as separate rule families. Algebraically we summarise them as a monotone

transformation of the normalised signal that pushes **only upward** (worsening stability reading) when documented thresholds are met:

[\hat{x}{c,k,t} \;=\; \min!\Bigl(100,\; \mu^{\mathrm{con}}{c,t}\,\cdot\,\mu^{\mathrm{frag}}{c,t}\, \cdot\,\tilde{x}{c,k,t}\Bigr)\,.]

Note: Concrete **triggers** for (\mu^{\cdot\,\cdot}) (e.g. episodic conflict indicators vs. structural fragility) must be kept **separate** in layer text so audits do not conflate **semantics**.

Weighted end composition:

[S{c,t}^{\mathrm{raw}} \;=\; \frac{\sum{k\in\mathcal{K}{c,t}} \omega{c,k,t}\,\hat{x}{c,k,t}} {\sum{k\in\mathcal{K}{c,t}} \omega{c,k,t}} \qquad\text{(if denominator }>0\text{; else documented “empty country-day” path).}]

Upper bound:

[S{c,t} \;=\; \min\bigl(S{c,t}^{\mathrm{raw}},\, U_c\bigr)\,.]

Virtual currency nodes (monograph) are modelled as **special** connectors carrying raw FX rates but entering only via defined **country anchors** in (\mathcal{K}_{c,t}) — not as “global” connectors without territorial reference.

Pseudocode (compact)

```
function stage3_country_score(c, t, Configuration K): K_active ←
filter_country_nonstatic(Connectors, c, t) numerator ← 0; denominator ← 0 for k in
K_active: w_eff ← normalised_base_weight(k, K) * population_scale(c,k,t) *
institutional_coupling(c,k,t) if contribution_missing(c,k,t) per K: x_tilde ←
K.substitution_level_eta else: x_tilde ← stage2_output(c,k,t) x_hat ←
apply_conflict_fragility_malus(x_tilde, c, t, K) // monotone, documented, cap at 100
numerator ← numerator + w_eff * x_hat denominator ← denominator + w_eff if denominator
== 0: return documented_empty_day(c, t) // status/vintage/required fields s_raw ←
numerator / denominator return min(s_raw, K.cap_U[c])
```

This pseudocode **does not replace** mandatory publication of concrete numeric values for (w_k), (\eta), or (\mu); it defines the **accountability frame**: every change to weights or malus triggers must be **traceable** (version, date, reference to monograph/register).

[Figure 3a: Stage 3 — weight graph: (w_k) → (\pi,\iota) → substitution/malus → weighted sum → cap]

3.5 Relation to verbal specification and supplementary archive

Two-track publication practice remains: (i) this whitepaper delivers the **publicly citable** reconstruction; (ii) **numeric** weight tables and machine-readable policy artefacts may be provided in a **supplementary archive** when cleared. Until then, stages 1–3 remain a **rulebook** operationalised in layer documentation and source registers.

3.6 Relation to engine and LPU inference

Where **Naciro** and **LPU** are described in the Knowledge Graph, this denotes **inference and throughput logic** for documented transformations and delivered fields — not replacing stages 1–3 with an undocumented end-to-end AI. Rather, **rule-based** and **inference-assisted** components are positioned **along the data path**; the NFSI remains bound to **transparency of end aggregation** [^methodik].

3.7 Sensitivity to information asymmetry

Media and connector landscapes are globally **uneven in density**. The pipeline must not enforce an implicit equation "absence of news = stability"; recovery and starting-value logics in stage 2 and confidence and vintage displays in macro surfaces are **necessary** correctives (NationFiles Research, 2026, Parts C.5, W.3d) [^monographie].

[Figure 3: Data flow — from heterogeneous connectors through stages 1–3 to NFSI country headline and derived world aggregates]

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4. PRESENTATION ONTOLOGIES AND AUDIENCES

The monograph explains why **many controllers** exist: each analytical audience needs its own **presentation ontology** without duplicating the data basis (NationFiles Research, 2026, Part F) [^monographie]. Table 1 summarises the ontology inventory (Part J) [^monographie].

Table 1. Excerpt from presentation ontology inventory (simplified).

ONTOLOGY	PURPOSE	TYPICAL AUDIENCE ROLE
World situation overview	Headline values, global framing	Public, media
Country depth	NFSI layers, subsites, news	Analysts, NGOs
Comparison pair	Side-by-side, fair vintage notes	Macro, policy
Security board	Filter lenses, hotspots, export	Security, OSINT
Macroeconomics (PPI)	Rankings, choropleths, scatter plots	Economists
Governance (GGI)	Institution metrics	Policy, reform advisory
Legal / source ontology	Provenance, connector register	Compliance, science
Knowledge graph	Definitions, edges, mind maps	Editorial, research
Export & badge	Micro-citation	Technical partners

Knowledge-graph terms (NFSI, Engine, LPU, entity families) stabilise **semantic translation** between internal pipeline and public explanation. Where graph definition and SEO structured data diverge, **harmonisation** or clear derivation is required — otherwise parallel "truths" emerge that undermine trust in a **persistently identified** publication (Part W.1d) [^monographie].

4.1 Dashboard and global world situation (C.2)

The entry layer is designed as a **synthesis layer**: world map with stability colouring, aggregated world index, 30-day world-index time series, localisable chart strings, plus embedded short news and event windows. Export paths deliver the **same series** in machine-readable form — an integrity-preserving pattern against screen-scraping (NationFiles Research, 2026, Part C.2) [^monographie].

[Figure 4a: Dashboard data flows — map, time series, news, status export]

4.2 Country domain as multi-subsite system (C.3)

The country domain bundles **news**, **metadata**, **metamaps**, **security radar**, **travel**, **migration**, **country comparison**, **NFSI detail**, **short-horizon windows**, **snapshots**, and **export PDFs**. Canonical and translation discipline ensure **mobile short profiles** and **desktop dashboards** show the same canon values (NationFiles Research, 2026, Part C.3) [^monographie].

4.3 Map and economics controllers (C.4–C.6)

Map controllers unite **hub logic**, thematic metamaps, and security-related world maps (including travel advisories, earthquakes, short-horizon military/protest windows). Economics controllers implement **PPI** and **GGI** layers with **metric registers**, confidence codes, and audit-friendly tooltips — deliberately **not** identical to NFSI (NationFiles Research, 2026, Parts C.4, C.5) [^monographie].

4.4 Security, law, knowledge, and export (C.7–C.11)

Security controllers combine **global radar** (filter lenses, export) and **wanted-person consolidation** of sensitive data with strict **404 discipline**. Law controllers expose **layers**, **registers**, and full-text search. Knowledge controllers stabilise **entities**, **FAQ**, **graph mind maps**, and export bundles. Export controllers enable **badges**, **feeds**, and machine-readable artefacts (NationFiles Research, 2026, Parts C.7–C.11) [^monographie].

[Figure 4: Projection — one relational truth into multiple controller ontologies; example paths dashboard vs. country depth vs. export]

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5. VALIDATION, STRESSORS, AND DATA INTEGRITY

The internal monograph develops **auditable case studies** and review catalogues (Part O, extended by Q–U) readable as **methodological stress tests**: election weeks, sanction shocks, territorial conflicts, earthquake layers over daily NFSI, merged lists of sensitive domains, multilingualism, accessibility, and PDF archivability.

Core thesis of this chapter: Integrity arises not only from technical availability but from **made-visible assumptions** (vintage, confidence, gatekeepers) and from the ability to **defuse misreadings** through text, legend, and status discipline.

5.1 Validation logic: What a "stress test" means here

Unlike classical ML benchmarks, stress tests **do not** target a single loss value but **epistemic robustness**: Does the **reading** of data stay stable in politically volatile weeks? Do **false reassurances** arise from cache, from combining maps of different temporal resolution, or from diverging structured metadata?

5.2 Example fault trees (excerpt)

- **Connector outage:** neutral substitution in stage 3 — **meta situation** must name domain.
- **Geometry outage:** textual fallbacks, no silent blank maps.
- **Timestamp drift:** macro vintage vs. NFSI as-of date separately shown.
- **Gatekeeper misclassification:** escalation path instead of algorithmic closure alone (NationFiles Research, 2026, Parts H, O) [^monographie].

[Figure 5: Example fault tree "mid-run connector outage" — fallback, communication, meta situation]

5.3 From review catalogue to case study: methodological frame

Internal catalogue IDs (Part O, incl. O.75, O.8, O.36, O.55) [^monographie] are **not** empirical measurement series but **scenario anchors**. For a persistently identifiable paper we reconstruct **fictional yet realistic** time paths: they illustrate **which observables** (signal density, connector availability, separate macro vs. NFSI vintage) must be visible for audit and review. All numbers in Tables 2–4 are **illustrative** for **didactic** readability, not claimed proof of a historical event.

5.4 Case study A — Information shutdown and signal-density collapse (ref. O.75)

Setup (fictional): In **"Demokratia,"** a **wide-ranging Internet shutdown** occurs between (t{=}0) and (t{=}14). OSINT connectors relying on public news pillars and civil-society sources lose **observability**, while still-reachable satellite/bank/commodity paths **partly** continue.

Pipeline expectation: The NFSI architecture must **not** infer "calm" automatically from **missing** headlines. Recovery and substitution logic in stages 2–3 must yield either (a) a **conservative** country score or (b) **uncertainty/confidence bands** and status fields surfacing the **information vacuum** — per released policy fixed in layer text.

Table 2 shows a **qualitative** path (0–100 scale illustrative only for "normalised stress reading").

Table 2. *Fictional* day indicators under information shutdown.

DAY (T)	PUBLIC NEWS SIGNAL DENSITY (INDEX)	SHARE OF AVAILABLE OSINT CONNECTORS	ILLUSTRATIVE RAW STAGE-2 AGGREGATE INPUT	COMMENT
−2	62	0.94	54	Baseline
0	58	0.91	56	Start of restriction
3	22	0.61	59	Echo collapse — *without* ethical recovery, "silence = good" would be conceivable
7	11	0.38	61	Vacuum — pipeline must flag gaps
10	9	0.33	58	First partial routing workarounds
14	18	0.45	55	Recovery of observability

[Figure 12: Qualitative curves — signal density vs. stage-2 raw aggregates vs. policy-dependent NFSI path with confidence band (placeholder)]

Audit questions (from O.75): Is the **information vacuum** named semantically on the country surface? Does substitution-plus-malus prevent **artificial calming** while uncertainty remains high?

5.5 Case study B — Sanction shock with diverging macro paths (ref. O.8, O.55)

Setup (fictional): "Handelsrepublik" experiences a **sanction shock** at (t{=}0). Commodity and FX connectors jump; **PPI-related** series react **fast**, **GGI/governance** series **slowly**. NFSI must **not** coincide with a single FX spiral.

Table 3. *Fictional* separated paths (0–100, higher = greater stress in respective domain reading).

DAY	NFSI-ALIGNED DOMAINS (COMBINED)	PPI STRESS PROXY	GGI INSTITUTIONS PROXY	NOTE
−5	48	41	44	Pre-shock
0	53	68	45	Shock day — FX/ commodity steep

DAY	NFSI-ALIGNED DOMAINS (COMBINED)	PPI STRESS PROXY	GGI INSTITUTIONS PROXY	NOTE
5	56	71	46	PPI "hot," GGI barely moves
14	58	64	49	partial market adjustment
30	57	59	52	institutional lag visible

[Figure 13: Triple time series — NFSI vs. PPI vs. GGI; mandatory vintage note per series (placeholder)]

Audit questions: Side-by-side **comparison** of two countries must not suggest fair conclusions without symmetric **vintage** (O.20). **Dual-exchange-rate** or confidence scenarios (O.55) must be tooltip-explainable so NFSI is not misread as a **synonym** for exchange-rate policy.

5.6 Case study C — Recovery after data gaps vs. real volatility (ref. O.36)

Setup (fictional): A compute-heavy connector is **down for days**; ground truth remains **volatile**. Recovery rules smooth **gaps** but must not suggest the situation is "already normalised" when external observers still see escalation reports.

Table 4. *Fictional* interaction of gap + recovery.

PHASE	EXTERNAL CRISIS SCALE (EXPERT POLL, FICTIONAL)	INTERNAL GAP FLAG	NFSI SCENARIO A (OVER-OPTIMISTIC RECOVERY)	NFSI SCENARIO B (CONSERVATIVE + VISIBLE UNCERTAINTY)
A	high	no gap	58	58
B	high	gap active	52 ← **suspicious**	61 ← consistent with volatility
C	medium	gaps closing	55	57

Scenario A is **methodologically unacceptable** if produced by **default substitution**; it serves as a **negative** teaching example. Scenario B shows the **integrity path**: higher or explicitly band-based values while uncertainty and gaps coexist (cf. integrity strategy Part W) [^monographie].

6. DISCUSSION: GOVERNANCE, ETHICS, AND PLATFORM CREDIBILITY

6.1 Integrity strategy (Part W)

We summarise the **integrity strategy** as follows (fully developed in NationFiles Research, 2026, Part W) [^monographie]:

- **Avoid** empty geopolitical promises in navigation;
- **Restraint** toward cartographic pseudo-precision;
- **Reduction** of static prose without data tie-in in favour of data-driven, versioned artefacts;
- **Unification** of diverging structured-data branches;
- **Synchronisation** of pipeline changes with **persistently identifiable** methodology publications;
- **Mobile UX** as its own **speed class** with immediate readability of headline indicators;
- **Descriptive KPI language** instead of moralising shorthand;
- **Higher frequency** of honest status and freshness reports supporting a **continuous situational picture**.

The principle **"transparency over elegance"** is thus not aesthetic but **epistemic**: smooth surfaces hiding uncertainty harm trust even when they feel more "convincing" short term.

6.2 Global South and information asymmetry

Where media or connector coverage is thin, the platform must surface **information asymmetry** — so absence of headlines is not misread as stability (Part W.3d) [^monographie].

6.3 Data sovereignty and rights of Indigenous peoples and communities

Governance of a global OSINT and macro framework touches **not only** state sovereignty in the narrow sense but **epistemic justice**: many Indigenous peoples and locally rooted communities are **under-represented** or **distorted** in public and commercial data ecosystems — e.g. when territories appear only as aggregated national area, when resource conflicts lack land-rights perspective, or when language bias in news and event corpora amplifies dominant narratives (review logic related to O.7 subnationality, O.5 source bubbles) [^monographie].

We summarise **operational guardrails** coherent with the integrity strategy:

1. **Territorial and colonial caution:** Where the monograph treats subnationality and autonomous regions (Part O.7), later **finer** spatial modelling (Part W.3 — subnational extension) must be paired with explicit **legal and ethics review**, rather than silently homogenising Indigenous land claims under state surfaces.
2. **Provenance dominance:** For sensitive topics (land, resources, health, religion), **source and register discipline** takes precedence over "story optimisation" — **transparency over elegance**.

3. **Voluntary data and community-in-the-loop:** Where possible, **consultative** validation windows and documented objections in status or methodology artefacts — not a substitute for democratic representation but a **safeguard** against mono-causal external attribution.
4. **GDPR and DPIA paths:** Pseudonymisation and personal-data traces (O.60) remain **mandatory**; a geopolitical index must **not** smuggle in covert surveillance legitimacy.

These guardrails **do not replace** international-law or ethnological expertise; they mark the **method bridge** between internal pipeline and normative expectations from agencies and research on **Indigenous data sovereignty** debates (self-governed data, CARE principles; cited here only as **external** orientation, not an exhaustive bibliography).

6.4 Legal and security-policy limits

The NFSI and related visualisations **do not replace** consular or military decisions, jurisdiction, or sanction interpretation. Their role is **informational and rule-based**. Disclaiming wording in travel and security contexts must be **multilingually** consistent.

6.5 Scientific citation ethics

When NFSI is cited in policy papers, the **referenced methodology publication** (ideally via a persistent identifier) should be preferred to bare URL citation. **As-of date** and **language version** should accompany citations, as surface text can change faster than pipeline logic.

[Figure 10: Governance stack — layer documentation, registers, Knowledge Graph, integrity strategy Part W]

7. CONCLUSION AND OUTLOOK

We sketch a path for NationFiles to function as **open-science-compatible citation infrastructure**: the **three-stage pipeline** as explainable core, plural presentation ontologies as **audience-specific projections** of one data basis, and an **integrity strategy** that pushes back pseudo-precision and semantic double tracks.

Outlook (strategic fields, Part W.3): **subnational modelling**, **peer-review layers** on the knowledge graph, **climate–migration ontology** strictly separated from short-horizon NFSI tactics, plus inclusion and audit mechanisms — each only with **governance and maintenance budget** so new layers do not become empty gestures.

7.1 One-sentence summary for citation purposes

NationFiles materialises a **documented rule-based** NFSI through a **three-stage pipeline** from OSINT and macro signals and projects identical headline metrics into **several professionally grounded presentation ontologies**, supported by a **public Knowledge Graph** and an **integrity strategy** that prefers transparency to cosmetic smoothness.

APPENDIX A — DEEPENING THE METHODOLOGY (READINGS OF THE THREE-STAGE PIPELINE)

A.1 Epistemic function of Stage 1

Stage 1 answers how **heterogeneous raw signals from one source** translate into **comparable metric language**. Directional sensitivity prevents false domain transfers between e.g. economically optimistic and security-relevant readings.

A.2 Temporal inertia and recovery in Stage 2

The mix of **today and yesterday** dampens single-day outliers. Recovery rules for missing days are **ethically** salient: missing data must not silently be read as normality.

A.2.1 Algebraic sketch (day aggregation and recovery)

Let (y_{c,k,t}) be the normalised connector contribution from stage 1. A **minimal** reconstruction of the "today–yesterday" mix is:

[x{c,k,t} \;=\; \alpha\, y{c,k,t} \;+\; (1-\alpha)\, y_{c,k,t-1}\,, \qquad \alpha \in (0,1)\text{ from documented policy.}]

Recovery: If (y_{c,k,t}) is missing, a **gap function** (R(\cdot)) applies — e.g. limited carry-forward, cap against over-smoothing, or explicit "uncertainty flags":

[x{c,k,t} \;=\; R\bigl(y{c,k,t-1},\,y_{c,k,t-2},\,\ldots;\,\text{Policy}\bigr)\,.]

(R) must **not** arbitrarily push the aggregate reading downward when external crisis indicators remain high — cf. case study 5.6.

A.3 Weighting and institutional coupling in Stage 3

Stage 3 is where distinct **connector families fuse** under documented weights. **Transparency of substitution** for missing contributions is mandatory for citation and agency communication.

A.4 Distinguishing NFSI ↔ prognosis

The NFSI is **descriptive and rule-based**. Prognostic product components — if delivered — must be **separately** named, dated, and versioned.

[Figure 6: Stage-wise information-algebraic reading — what each stage may claim and what it may not]

APPENDIX B — CATALOGUE OF EXEMPLARY INTEGRITY STRESS TESTS (EXCERPT PART O)

ID	SCENARIO	CENTRAL AUDIT QUESTION	EXPECTATION
O.1	Election week	Are news and NFSI time logic visibly separate?	No index “judgment on elections”
O.3	Territorial conflict	Is territorial fallback explained?	No silent international-law vacuum
O.4	Earthquake + NFSI	Is non-causality visible?	No narrative shortcuts
O.8	Macro vs. NFSI	Are institutional tooltips present?	No “rich = stable”
O. 12	NGO PDF deadline	PDF with stamp/language complete?	Archivability
O. 39	Gatekeeper false negative	Escalation path exists?	Human-rights situation
O. 42	Structured data vs. live	Fields match deployment?	One version truth

[Figure 7: Stress-test workflow — scenario → UI/meta situation → documentation]

APPENDIX C — PRE-PUBLICATION CHECKLIST (EDITORIAL)

1. German and English titles aligned with the released artefact; 2. full author line incl. affiliation; 3. document type and version date; 4. abstract identical to print/PDF output; 5. keywords; 6. CC BY 4.0 licence visible; 7. linked identifiers (parallel publication, source code, graph) only after coordination; 8. PDF and optional source markdown of the same version; 9. on pipeline changes, update methodology text and version number (cf. monograph Part W.2a) [^monographie].

[Figure 8: Metadata flow — manuscript → repository → profiles]

APPENDIX D — MIRRORS AND SECONDARY PUBLICATION

Uniform citation: Summary, year, and licence must match the **canonical** publication artefact; distributed copies (repositories, academic profiles) must not carry diverging abstracts without a version note.

[Figure 9: Reference graph — canonical artefact as root]

APPENDIX E — BIBTEX ENTRY

Enter persistent identifier (`doi` or equivalent) **after assignment**; until then omit or comment out.

```
@techreport{neawolf2026algorithmicgeo, author = {Neawolf, Sven}, title = {Algorithmic Geopolitics: Methodology of AI-Driven Real-Time Stability Indexing within the NationFiles Framework}, institution = {Neawolf Media Group}, year = {2026}, month = apr, note = {Technical Whitepaper v1.0. German parallel title: Algorithmische Geopolitik: Methodik der KI-gestützten Echtzeit-Stabilitätsindizierung im NationFiles-Framework. Persistent identifier to be added upon publication.} }
```

APPENDIX F — EXTENDED REVIEW CATALOGUE (CONDENSED FROM PART O)

The following list extends Appendix B with further **typical crisis and operations scenarios**. It is **not** an exhaustive test matrix but a **working set** for QA and second readers (NationFiles Research, 2026, Part O) [^monographie].

ID	SHORT NAME	AUDIT CORE
O.5	Source bubbles	diversity vs. echo
O.6	Migration time series	year/definition visible
O.7	Autonomous region	subnationality
O.10	Wanted-person inconsistency	source per field set
O.14	Bot traffic	news robustness
O.15	Dark-mode badge	contrast
O.20	Compare vintage	PDF fairness
O.24	Analytics vs. situation	firewall
O.28	Tooltip cache	TTL visible
O.31	Whitepaper drift	versioning
O.35	Ethics board pause	logging duty

ID	SHORT NAME	AUDIT CORE
O.44	Rare earth macro	semantics ≠ NFSI
O.55	Dual exchange rate	confidence
O.60	DPIA pseudonyms	privacy review
O.65	Travel-AR tone	legal form
O.70	Sea-route meta	no operational how-to
O.75	Internet shutdown	information vacuum

[Figure 11: Heatmap — scenarios × subsystems (qualitative, placeholder)]

F.1 Long-form contextualisation (reading depth)

Redundancy between main text and appendices is **intentional**: persistently identified publications are often read **linearly**. Stage 1 establishes **source loyalty** — each source has its own "metric grammar" before entering the common scale. Stage 2 models **country time** — the index behaves not like a raw ticker but like a **smoothed yet responsive** signal. Stage 3 makes **weighting policy** visible. This triple structure makes the NFSI **debatible** — a prerequisite for **agency understanding** and **academic critique**.

F.2 Addendum on presentation ontology

Controllers are **epistemic gateways**: country controller **depth**, dashboard **global coherence**, economics controller **macro–governance separation**, security controller **dual lens** between event situation and normalised person search.

F.3 Addendum on integrity strategy (Part W)

Transparency over elegance is a **publication principle**: flag ambiguity honestly; avoid empty navigation promises; harmonise diverging structured data so only **one** authoritative public definition remains.

APPENDIX G — BRIEF NOTE ON MIRRORING CONTENT

After first publication, **abstract, year, and licence** of all publicly visible copies should match the canonical artefact. Technical distribution channels (repositories, profile pages) are **secondary** to the **methodological core** of this document and may change; the argumentative version here is authoritative.

ACKNOWLEDGEMENTS AND CONFLICT OF INTEREST

Acknowledgements: to all specialist and operations teams enabling operational integrity and documentation discipline.

Conflict of interest: The author holds responsibility in the organisation operating NationFiles; methodological claims refer to **documented** public artefacts (Knowledge Graph, layer texts, registers), not to unverifiable marketing claims.

REFERENCES AND SOURCES

[^monographie]: NationFiles Research (2026). *NationFiles — Geopolitisches System: Ausführliche Gesamtbeschreibung (Backend, operative Datenpipeline und Frontend).* Internal system and specialist monograph, `built_at_utc` 2026-04-30. **Primary source** for architecture and controller landscape.

[^methodik]: NationFiles Research (2026). *Methodik und Anwendung der KI-gestützten geopolitischen Risikoanalyse: Das NationFiles Framework und die Naciro Intelligence Engine* (and English-language parallel edition). Cite the valid specialist reference artefact as of first publication.

Profile and repository pages should use the **same** summary, year, and licence as the canonical publication artefact.

LIST OF FIGURES (PLACEHOLDERS)

1. Paradigm shift archive → continuous situational picture
2. Architecture overview backend/frontend/knowledge
3. Data flow stages 1–3 to NFSI
   3a. Stage 3 — weight graph (substitution, malus, cap)
4. Ontology projection and example paths
   4a. Dashboard data flows (map, time series, export)
5. Example fault tree / integrity audit
6. Stage guardrails (epistemic claim limits)
7. Stress-test workflow
8. Metadata flow — manuscript to profiles
9. Reference graph canonical artefact
10. Governance stack
11. Scenario heatmap (review catalogue, placeholder)

12. Case study information shutdown — quality curves (fictional)
13. Case study sanction shock — NFSI vs. PPI vs. GGI (fictional)

NOTE ON EXPECTED PDF PAGE COUNT

In the in-house research PDF pipeline (current state after adding Stage 3 algebra, case studies, and governance deep dive), this manuscript typically yields **about 22–35 printed pages** — depending on font size and hyphenation. For a **40–50 page** target, consider (i) **full** algebraic specification also for stages 1–2, (ii) **large** tables from monograph Parts O/Q, or (iii) larger base font / line spacing in the PDF profile (`ResearchPdfBuilder`, etc.); two-column layout only after validating the reference pipeline.

End of whitepaper manuscript — Version 1.0, 2026-04-30.

As of: 2026-04-30

Only the German version is legally binding.

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nationfiles.com | Algorithmic Geopolitics: Methodology of AI-Driven Real-Time Stability Indexing within the NationFiles Framework | 2026-04-30 11:59:20 UTC

NATIONFILES STABILITY INDEX (NFSI) - VALIDATION AND VERIFICATION REPORT

NationFiles Research

STAND 2026-04-26 · EN

NationFiles

Neawolf Media Group

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NationFiles Stability Index (NFSI) - Validation and Verification Report

As of: 2026-04-26

PDF from: 2026-04-26 07:18:00 UTC

ABSTRACT

This report describes the methodology, data provenance, and technical traceability of the NationFiles Stability Index (NFSI) and the 7-day forecast. It is intended for universities, public authorities, and accredited auditors to verify and endorse the credibility and reproducibility of the system. NationFiles is committed to absolute transparency: all formulas and parameters stated correspond to the actual implementation; no values are invented or adjusted ex post. The NFSI is suitable as a reliable reference for scientific studies and audits.

SUMMARY

The NFSI is based on a three-layer aggregation (Layer 1: indicators, Layer 2: thematic smoothing with 7-day inertia, Layer 3: weighted final score including crash mode). All weights and formulas are fixed and implemented accordingly.

Data are drawn from public and licensed sources (e.g. World Bank, ACLED, INTERPOL, GDELT); full provenance and licenses are available in the data inventory and at `/legal/sources/`.

The 7-day forecast uses a VAR-based method with 90-day history and multiple time series (e.g. world index, country-specific news risk). The methodology is consistently documented here and in the implementation.

Governance includes version control, audit trail, and defined processes for external auditors.

SCOPE AND TARGET AUDIENCE

Target audience: universities, research institutions, national and international authorities, and accredited auditors who are to verify or endorse the validation and recognition of the NFSI and the forecast. Geographic coverage: global. Temporal coverage of the NFSI data basis: from February 2026 onwards, ongoing. Not in scope: individual investment or financial advice; the NFSI serves to assess stability and risk, not to recommend financial products.

INTEGRITY AND CREDIBILITY OF THE NFSI

NationFiles and the NFSI are designed to serve as a trusted reference in scientific studies, audits, and regulatory assessments. The following positions underscore traceability and defensibility.

- **No invented values:** All NFSI values and forecasts are produced solely from documented formulas and the sources listed in this report and the data inventory. There are no manual overrides or black-box corrections.
- **Fully code-based and documented:** Every calculation step (Layers 1–4) is implemented in code and described in this report with constants, formulas, and pseudo-code.
- **Provenance and audit trail:** Data origin, licences, and update frequencies are documented in the data inventory and legal sources. Changes to model and data are versioned and traceable via an audit trail.
- **Governance** includes version control, audit trail, and defined processes for external auditors.
- **Algorithmic interpretations** do not represent official assessments. Technical security, audit cycles (e.g. SCA/SAST), and infrastructure are described in the Global Security & Resilience Statement (`/legal/security/`).
- **Error correction for raw data:** report to data@nationfiles.com → source review → correction in the database → automatic recalculation; manual overwriting of scores is excluded (see Independence Statement `/legal/independence/`).

ABSOLUTE TRANSPARENCY

NationFiles is committed to full transparency in the methodology, data, and calculation of the NFSI. There are no hidden parameters, no opaque corrections, and no deliberately withheld information relevant to assessing the index.

What is disclosed:

- All formulas and calculation steps (Layers 1–4), including pseudo-code and the complete constants table, in this report.
- Complete data inventory of all sources with NFSI impact: origin, licence, update frequency, thematic weight. Access to legal sources (`/legal/sources/`).
- No secret weightings: The weight matrix (connector weights, groups) is documented; upon request, authorised auditors can access technical artefacts under audit agreements.
- Changes to model and data are versioned; an audit trail (e.g. timestamps, hashes) ensures traceability. Calculation logs per country/date can be provided during audits to verify intermediate steps.

- No black box: Every published NFSI value can be reconstructed from the documented formulas, constants, and input data. There are no downstream "secret corrections" or manual overrides.

Transparency is not a marketing claim but documented practice: this report and the linked references are the basis.

HOW TO CITE THIS REPORT

This report is citable for media, academia, and public authorities. Use the following:

> *Neawolf Media Group / NationFiles (2026). Validation and Verification Report — NationFiles Stability Index (NFSI). Version 1.0. Aachen. Document ID: VVR-2026-03-07.*

Short:

> *NationFiles (2026). NFSI Validation Report. VVR-2026-03-07.*

SYSTEM ARCHITECTURE AND DATA PIPELINE

Data ingestion → Preprocessing → Layer 1 (indicators, normalization) → Layer 2 (thematic aggregation, 7-day smoothing) → Layer 3 (final score, crash mode) → Delivery via API and feeds.

Fig.: Data flow from ingestion through Layers 1–3 to delivery.

1. Ingestion
2. Preprocessing
3. Layer 1 — Indicators
4. Layer 2 — Thematic aggregation
5. Layer 3 — Final score and crash mode
6. 7-day forecast / API

WEIGHTING, SCALES, AND NORMALIZATION — HOW EVERYTHING IS WEIGHTED

So that the reader can see unambiguously how the system works: all value ranges, normalization rules, and weightings are described here in one place.

Value ranges (scales) at a glance

All stability scores in the NFSI system lie in the interval `[0, 100]` after processing. Higher = more stable (better), lower = less stable (worse). Exception: raw values before Layer 1 are source-specific and are mapped to 0–100 in Layer 1.

QUANTITY	RANGE	MEANING
Raw value (raw)	source-specific	Original value from the data source (e.g. count, percentage, Goldstein −10 to +10). Normalized in Layer 1.
score_row (row score)	[0, 100]	After Layer 1: one score per row (e.g. per country/date/event). Default 50 when spread is zero or min/max missing.
dayScore (daily score per connector/country)	[0, 100]	After Layer 2 aggregation: one value per connector, country, and date. Security group: MIN of row scores; others: average (with dummies 0, 100 and pad 50).
score_final (Layer 2 output)	[0, 100]	`0.6 × dayScore + 0.4 × previous day`; after recovery max 95. Stored in nfsi_connector.
baseScore (Layer 3)	[0, 100]	Weighted average of all connector scores (`effW = group × scoreValue/100 × updateMult`). Dummies 0 and 100 with weight 1.
rawScore / nfsi_raw	[1, 100]	After conflict malus, fragility malus, small-country malus, population bonus, WGI pull. Floor 1, cap 100.
NFSI_Today (final index)	[1, 100]	After Layer 4 (inertia): daily change capped at ±3; in security crisis (`min security < 25`) immediate pass-through without smoothing.

Layer 1 — Normalization and direction

Per data source, one raw-value column (or a computed severity) is used. Over all rows of that source, minimum (`min_raw`) and maximum (`max_raw`) are computed; `span = max_raw − min_raw`. If `span ≤ 0` or min/max not defined, all rows receive the neutral `score_row = 50`. Otherwise: per row `normalized = (raw − min_raw) / span × 100`, i.e. values in `[0, 100]`.

Direction is fixed per indicator:

- “higher raw = worse stability” (e.g. conflict count) → `score_row = 100 − normalized`
- else `score_row = normalized`

The result is clamped to `[0, 100]` and rounded to two decimals. Thus every row has a uniformly interpretable score: 0 = worst, 100 = best stability within that source.

Layer 2 — Today vs. previous day weight and aggregation rules

Per connector, country, and date, all associated row scores (`score_row`) are collected.

- **Security group (group 100):** daily score is the minimum of collected scores; missing values are padded with 100 (conservative: one critical event suffices).
- **All other groups:** fixed dummy values 0 and 100 plus pad value 50 for missing entries are used so that all countries have the same effective array length; daily score is the arithmetic mean of this array.

Then smoothing with previous day: `score_final = 0.6 × dayScore + 0.4 × yesterday` (yesterday: last available value; if missing, 70 for security, 85 for others). Value is always in `[0, 100]` .

- **Recovery:** if a day is missing, score is increased by 0.2 (security) or 1.0 (others) per day, up to 95, for at most 90 days.

Layer 3 — Connector weights, groups, and effective weight

Each connector has a configured weight (e.g. 40–100, see data inventory) and a thematic group (e.g. 40, 50, 60, 85, 100; group 100 = security). The effective weight in the weighted average is: `effW_n = group_n × (scoreValue_n/100) × updateMult_n` where `scoreValue_n` is the connector weight/100 and `updateMult_n` reflects update frequency (e.g. 1.0 annual to 7.3 multiple times daily).

Missing connector data (no_data) receive the neutral score 50. The weighted average is formed with fixed dummies 0 and 100 (weight 1 each) → `baseScore` .

Then maluses/bonuses apply:

- conflict malus (if `min(security) < 70`)
- fragility malus (`governance gap × population sensitivity`)
- small-country malus (if `pop < 5M`)
- population bonus ($\log_{10}(pop)\times 0.5$, max 4)
- WGI pull (governance 0–100 raises raw score)

All intermediate results are clamped to `[0, 100]` ; final raw score has floor 1 and cap 100.

Layer 4 — Inertia weights and daily change cap

The raw score from Layer 3 (`nfsi_raw`) is smoothed with the previous day's NFSI (`prevScore`) unless there is a security crisis (`min(security) ≥ 25`).

- **Inertia weights:** standard 80% previous day, 20% today (`inertiaWeight = 0.80`).

- When many connectors are missing (no_data share ≥ 50%), 45% previous / 55% today (0.45) is used.
- When "score without L1/L2 data" 50% / 50% (0.50).

The maximum change from the previous day is ±3 points. In security crisis (`min(security) < 25`) smoothing is skipped: `NFSI_Today = nfsi_raw` (immediate pass-through).

SYSTEM FLOW — THE NFSI AS A READABLE DESCRIPTION

The following describes the full flow from raw data to the published NFSI value as continuous text. This account matches the implementation and serves as a scientifically readable reference.

Raw data are ingested per data source (connector) and preprocessed (imputation, time zones, deduplication). Each source provides rows with at least one raw-value column or a computed severity.

Layer 1 (Indicators): For each data source, the minimum and maximum of raw values over all rows are computed. If the span is zero or undefined, all rows receive the neutral row score 50. Otherwise, per row the raw value is normalized to `[0, 100]` : `normalized = (raw − min) / span × 100` . Depending on the indicator, "higher raw = worse stability" → row score = `100 − normalized` , else row score = `normalized` . The result is clamped to `[0, 100]` and rounded to two decimals. Thus every row has a uniform stability score.

Layer 2 (Daily score per connector and country): For each connector, country, and date, all associated row scores are collected. For the security group (group 100), the daily score is the minimum of these values (missing padded with 100). For all other groups, the arithmetic mean of the collected values plus fixed dummies 0 and 100 and pad 50 for missing entries is taken. The final Layer-2 score is `0.6 × daily score + 0.4 × previous day` (previous day: last available value; else 70 or 85). The result lies in `[0, 100]` . Missing days may be filled by recovery (step 0.2 or 1.0 per day, cap 95, max 90 days).

Layer 3 (Country NFSI): For each country and date, the Layer-2 scores of all connectors are available. Missing connectors are included with the neutral score 50 (no_data). Each connector's effective weight is `group × (connector weight/100) × update multiplier` . The base score is the weighted average of all scores including fixed dummies 0 and 100 (weight 1 each). If the minimum of security scores is below 70, a conflict malus (max 35) is subtracted. Then fragility malus (governance gap, smaller countries stronger), small-country malus for population below 5M, population bonus ($\log_{10}(pop)\times 0.5$, max 4), and WGI pull (governance 0–100 raises score, factor 0.95) are applied. The raw score is clamped to `[1, 100]` (floor 1, cap 100).

Layer 4 (Daily smoothing): The raw score is smoothed with the previous day's NFSI: by default 80% previous, 20% today; with many no_data 45% or 50% today. The change from the previous day is capped at ±3 points. If there is a security crisis (minimum of security scores < 25), no

smoothing is applied and the raw score is output as `NFSI_Today`. The result is the published NFSI value for that country and date.

DAILY INDICATORS AND THEIR IMPACT ON THE NFSI

The NFSI integrates daily-updated sources that reflect stability in near real time. These include news events (GDELT), media tone, and country-specific risk signals. They enter via defined indicators in Layer 1 and affect the daily value through Layers 2 and 3.

Relevant daily sources: (1) GDELT — Goldstein scale, average tone, and event risk level per country; (2) media sentiment; (3) proprietary news risk signals with 24h/48h windows.

A live signal is formed from averages of Goldstein, tone, and risk level. If average risk exceeds a threshold, a deduction is applied. The result may be smoothed with the established NFSI index (e.g. 40% live, 60% index) to limit noise.

Simplified logic (pseudo-code, not implementation-identical): LIVE_BASE = 50 FOR each source: LIVE_BASE += (Goldstein_avg × k1) + (Tone_avg × k2) + (Sentiment_avg × k2) IF risk_avg > threshold THEN LIVE_BASE −= (risk_avg − threshold) × k3 LIVE_BASE = CLAMP(LIVE_BASE, 0, 100) IF established index present THEN SMOOTHED = 0.4 × LIVE_BASE + 0.6 × index

DATA INVENTORY — ALL SOURCES WITH NFSI IMPACT

SOURCE	FIELD	LICENSE	UPDATE	COVERAGE	WEIGHT
AcledMonthAll	Crisis & Safety Radar (Level 1)	ACLED / filtered, aggregated for Naciro Intelligence	wöchentlich	Global	90
AcledMonthAllOverview	Crisis & Safety Radar (Level 1)	ACLED / filtered, aggregated for Naciro Intelligence	wöchentlich	Global	95
CountriesConflictUcdpGed	UCDP GED conflict events	Free for academic, commercial, governmental use (see UCDP)	täglich	Global	85
CountriesCurrencyFromFa	7-Tage-Trend Währung	intern	täglich	Global	65

SOURCE	FIELD	LICENSE	UPDATE	COVERAGE	WEIGHT
CountriesGdeltGlobRadar	GDELT global event data	Unlimited use for academic, commercial, governmental (see Terms of Use)	mehrmals täglich	Global	85
CountriesNetAbuseIpDb	Abuse IPs by country (AbuseIPDB)	AbuseIPDB Terms of Use	täglich	Global	70
CountriesNetGrpTrfcAnom	Traffic anomalies by location	Cloudflare Terms of Use	mehrmals täglich	Global	55
CountriesNetSpmBot	Botnet C&C by country (Spamhaus)	Spamhaus Terms & Conditions	täglich	Global	70
CountriesNetTrfcAnom	Traffic anomalies (Cloudflare Radar)	Cloudflare Terms of Use	mehrmals täglich	Global	70
CountriesPopPrisn100K	Prison population rate per 100k	CC BY / Our World in Data	wöchentlich	Global	50
CountriesVatRatesApiVer	VAT rates by country (APIVerve)	APIVerve Terms of Service	wöchentlich	Global	60
CountriesVatRatesVatLup	VAT rates EU (VAT Lookup)	Refer to source (EC data)	wöchentlich	Global	60
CountriesWb65Up	Population 65+ %	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	10
CountriesWbAccsZs	Access to electricity %	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	60
CountriesWbCcEst	Control of corruption (estimate)	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	85
CountriesWbCdrtIn	Crude death rate	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	30

SOURCE	FIELD	LICENSE	UPDATE	COVERAGE	WEIGHT
CountriesWbCpiTotlZg	Inflation, consumer prices %	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	75
CountriesWbFpCpiTotl	Inflation consumer prices % (WB)	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	20
CountriesWbLe00	Life expectancy at birth	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	45
CountriesWbMilTotl	Military personnel total	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	40
CountriesWbNetUser	Internet users %	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	40
CountriesWbPopDpnd	Age dependency ratio %	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	20
CountriesWbPopGrow	Population growth %	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	25
CountriesWbPovGini	Gini index (income inequality)	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	60
CountriesWbPsrcP5	Intentional homicides per 100k	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	100
CountriesWbResTotlCd	Total reserves (incl. gold) US$	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	25
CountriesWbRlEst	Rule of law (estimate)	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	80
CountriesWbUem1524	Youth unemployment (15-24) %	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	70

SOURCE	FIELD	LICENSE	UPDATE	COVERAGE	WEIGHT
CountriesWbUemTotl	Unemployment %	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	55
CountriesWbVaEst	Voice and accountability (est.)	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	50
CountriesWbWgiPvCcRl	WGI: stability, corruption, law	CC BY 4.0 / filtered, aggregated	wöchentlich	Global	85
CountryNewsRiskLevel	News Risk (24h, Carry-Forward)	CC BY-ND 4.0	täglich	Global	100
DisasterGdacsHumNat	GDACS disaster alerts	GDACS Terms of Use	mehrmals täglich	Global	100
EconFredHerCountScOvSc	Overall Economic Freedom (Heritage)	Refer to Source Link	wöchentlich	Global	60
EconFredHerCountScRanks	World/region rank (Heritage)	Refer to Source Link	wöchentlich	Global	60
EconFredHerCountScRights	Property Rights (Heritage)	Refer to Source Link	wöchentlich	Global	100
EconFredHerCountScTax	Tax & gov. spending (Heritage)	Refer to Source Link	wöchentlich	Global	60
EurostatStatArblsQ	Unemployment rate (monthly)	Refer to Source Link	wöchentlich	Global	60
EurostatStatTps	Population on 1 January	Refer to Source Link	wöchentlich	Global	40
GovFbiMw	FBI Most Wanted / fugitives	OpenSanctions / FBI data – Public Domain (DOJ)	wöchentlich	Global	80
GovIntPol	INTERPOL Red Notices / wanted	INTERPOL Terms of Use	wöchentlich	Global	85
NewsMediaStck	Intelligence Briefing &		täglich	Global	50

SOURCE	FIELD	LICENSE	UPDATE	COVERAGE	WEIGHT
	sentiment (MediaStack)	Refer to Service Agreement			
NfMilitaryMassViolence	NfsiMLMV actor-day stress (NFSI)	Internal processing for NationFiles Intelligence	mehrmals täglich	Global	100
NfNewsCountry24	Country24hNews	CC BY-ND 4.0	täglich	Global	70
NfRawNewsSignalsEventRisk	Raw News Signals Event Risk	CC BY-ND 4.0	täglich	Global	100
OecdSdmxGdpvAnnpct	GDP volume growth % (annual)	OECD Terms & Conditions	wöchentlich	Global	60
OWIDDemocracyIndex	Democracy Index (EIU)	CC BY / Our World in Data	wöchentlich	Global	70
SdgGlobalIndexSdsn	SDG overall index (SDSN ArcGIS)	Refer to Source Link	wöchentlich	Global	60
TravelAdvisoryUnified	Travel warnings (multi-authority)	Refer to Source Link	täglich	Global	75
TravelStateGov	U.S. travel advisories Level 4 (Do Not Travel)	U.S. Government Work	täglich	Global	80
TrvlWarnCaVoy	Travel advice Canada (Voyage)	Open Government Licence – Canada	wöchentlich	Global	75
TrvlWarnDe	Travel & security advice (AA)	Refer to Source Link	wöchentlich	Global	75
TrvlWarnDeWarn	Travel & security advice (AA) – warning only	Refer to Source Link	wöchentlich	Global	75
TrvlWarnUk	UK travel advice (GOV.UK)	Open Government Licence v2.0	täglich	Global	75

SOURCE	FIELD	LICENSE	UPDATE	COVERAGE	WEIGHT
WorldEarthQuakeUsgs	Earthquakes worldwide (USGS)	USGS Public Domain	mehrmals täglich	Global	100
WorldUnivDataSetIpsa	World Universities Dataset (IPSA)	Refer to repository	wöchentlich	Global	25

INDICATOR CATALOG (LAYER 1) — EXAMPLES

Selected examples including the most important daily indicators (GDELT, Media Stack, Naciro News):

UCDP GED

- Armed conflicts (UCDP GED)
- Number of armed conflict incidents (UCDP GED) per month; column: high.
- events/month · More events → lower stability.
- Min-Max per source.
- https://ucdp.uu.se/

GDELT Global Radar

- News tone and Goldstein scale
- GDELT provides per country and time window the Goldstein scale (−10 to +10) and average tone of coverage. Higher Goldstein = less conflict.
- Scale / aggregation · Higher value (positive) → better stability.
- Aggregated per country/day; positive direction.
- https://www.gdeltproject.org/

News Media Stack

- Media sentiment
- Aggregated sentiment from news sources (Media Stack API) per country or region. Sentiment typically −1 to +1; mapped to 0–100 in Layer 1.
- Sentiment (aggregated) · Negative sentiment → lower stability.
- Source-specific; higher raw = worse.
- https://mediastack.com/

Naciro News (24h / Event Risk)

- Country risk 24h and event risk
- Proprietary indicators: 24h risk level per country from Naciro News and event risk level from raw signals. Daily assessment of news and events.
- Risk level 0–100 · Higher risk level → lower stability.
- Internal; negative direction (higher = worse).
- `/legal/sources/`

The full indicator list comprises 80+ data sources with associated weights and groups; details on sources and licenses at `/legal/sources/`.

LAYER 1 — INDICATORS IN DETAIL

In Layer 1, a stability score between 0 and 100 is computed per data source and row. The method depends on indicator type (value-based or count-based).

- **Value-based indicators:** A raw-value column per source is used. Min and max are taken over all rows of that source. Per row: `normalized = (raw − min) / (max − min) × 100`. If "higher raw = worse stability", `score_row = 100 − normalized`, else `score_row = normalized`. Result clamped to `[0, 100]`.
- **Count-based indicators:** A severity (e.g. per capita) is computed per row. A reference population (e.g. 1 million) is used when needed. The score is mapped to 0–100 consistently.

Layer 1 numerical example (value-based, simplified): Source "Conflict events", country X: raw values 5, 10, 20 for three time points. Min = 5, max = 20, span = 15. Row 1: (10−5)/15×100 = 33.3 → with "higher = worse": `score_row = 100 − 33.3 = 66.7`. Row 2 (value 5): `score_row = 100`; row 3 (value 20): `score_row = 0`.

Pseudo-code (value-based, higher = worse): MIN_RAW = MIN(raw) over all rows of source MAX_RAW = MAX(raw) over all rows of source SPAN = MAX_RAW − MIN_RAW FOR each row: IF SPAN = 0 THEN score_row = 50 ELSE normalized = (raw − MIN_RAW) / SPAN × 100 score_row = 100 − normalized score_row = CLAMP(score_row, 0, 100)

LAYER 2 — THEMATIC AGGREGATION AND INERTIA

Layer 2 aggregates Layer-1 scores per source, country, and date into a daily score and smooths with the previous day. Per (source, country, date) all associated `score_row` values are collected. Security-related sources (group 100): `daily score = MIN(scores)`. Other groups: `daily score = average of scores`, with fixed dummy values 0 and 100 for stable weighting.

Final Layer-2 score: `score_final = 0.6 × daily_score + 0.4 × previous_day_score`. If no entry exists for the day, a recovery value (up to 95) is used so that gaps are not penalised indefinitely.

Pseudo-code: DAILY_SCORES = COLLECT score_row FOR (source, country, date) IF group = security THEN daily_score = MIN(DAILY_SCORES) ELSE daily_score = AVERAGE(DAILY_SCORES, with dummies 0 and 100) prev = LAST_SCORE(country, source, previous_day) score_final = 0.6 × daily_score + 0.4 × prev score_final = CLAMP(score_final, 0, 100)

LAYER 3 — FINAL SCORE, CRASH MODE, AND SMOOTHING

Layer 3 combines all source scores into the country NFSI. Each source has a thematic group and weight; effective weight depends on current score. `effective_weight = group_weight × (score_node / 100) × update_multiplier`. Groups are e.g. 100 (security), 85, 60, 50, 40; group −1 is excluded.

`NFSI_Base = SUM(score_node × effective_weight) / SUM(effective_weight)`, including fixed dummies 0 and 100 with weight 1.

If the minimum of security scores (group 100) < 70, a conflict malus is applied: `malus = (70 − min_security) × factor`, capped (e.g. 35). `NFSI_Base` is reduced by this malus.

Optional: strong governance (e.g. WGI est_total) can pull the raw score up: `final = raw + (100 − raw) × (est_total/100) × wgi_pull`.

Daily smoothing (inertia): If security minimum ≥ 25, `NFSI_Today = prev × 0.8 + today × 0.2`, with max daily change ±3. In acute security crisis (min < 25), smoothing is suspended so the score reacts immediately.

Pseudo-code (core): FOR each source: eff_w = group × (score_node/100) base = SUM(score × eff_w) / SUM(eff_w) including dummies 0, 100 min_sec = MIN(scores of security group) IF min_sec < 70 THEN malus = (70 − min_sec) × 1.0; base = base − malus raw = CLAMP(base, 0, 100) IF min_sec >= 25 THEN nfsi_today = 0.8×prev + 0.2×raw; daily change max ±3 ELSE nfsi_today = raw

Fig.: Core formulas of the three NFSI layers (pseudo-code level).

AGGREGATION AND WEIGHTING (LAYER 2)

Layer 1: severity from row (node-specific); `impact = severity^1.6 * 100`; `score_row = max(0, 100 − impact)`. FBI/INTERPOL: `severity *= min(1, REF_POP/population)`, REF_POP = 1,000,000. Layer 2: raw = `AVG(score_row)` per (connector, iso2, date). `Score_Node = raw*0.2 + AvgLast7*0.8`. No entry → 100. Layer 3: `effective_weight_n = group_weight(connector) * (scoreValue/100)`. Groups: G1=100, G2=85,

G3=60, G4=50, G5=40, G6=-1 (excluded). `Score_Country,weighted = SUM(Score_Node * effective_weight) / SUM(effective_weight)`. `NFSI_Base`; resilience; `NFSI_Today = NFSI_Base*0.5 + NFSI_Last7*0.5`.

WORKED NUMERICAL EXAMPLE (ANONYMISED)

For illustration without disclosing the full weight matrix: a country has three indicators with Layer-2 scores 72, 85, and 90 on a given date. Thematic groups are 100, 60, and 40 with full value 100.

- Step 1: Effective weights: 72×1 = 72, 85×0.6 = 51, 90×0.4 = 36. Sum of weights = 159.
- Step 2: Weighted average (without dummies): (72×72 + 85×51 + 90×36) / 159 ≈ 80.2. With dummies 0 and 100 the order of magnitude remains comparable.
- Step 3: Min(security) = 72 ≥ 70 → no conflict malus. `NFSI_Base` ≈ 80.2.
- Step 4: Previous day = 79. `NFSI_Today` = 0.5 × 80.2 + 0.5 × 79 = 79.6 (simplified inertia).

The actual implementation uses additional factors (population bonus, governance pull, exact inertia parameters). This example is for traceability, not full replication.

DETAILED MATHEMATICAL CALCULATIONS AND PROCEDURE

The following formulas and steps match the implemented NFSI logic. All constants are fixed. Notation: raw = raw value, score = value in `[0, 100]`, sums and weights as stated.

CENTRAL CONSTANTS (REFERENCE)

These values are used in the formulas:

- **Layer 1:** If no spread (min = max) or no min/max → default score = 50.
- **Layer 2:** Today weight = 0.6; previous-day weight = 0.4. Dummy values: 0 and 100 (weight 1 each). Pad value for missing entries (non-security): 50. Security group (group 100): pad missing with 100 (conservative). Recovery per day: security 0.2 points, others 1.0; cap 95. Start without history: security 70, others 85.
- **Layer 3:** Conflict threshold = 70; malus factor = 1.0; conflict malus cap 35. Reference population = 45M; pop-neg multiplier cap 2.0. Small-country threshold = 5M; malus per $\log_{10}$ step = 4.0; cap 25. Population bonus: factor 0.5, cap 4.0. WGI pull = 0.95. No-data score = 50. Fragility: WGI scale 0–10, factor 3.0, cap 15. Score floor = 1.

- **Layer 4 (Inertia):** Standard 80% previous day, 20% today; with many missing data 45% or 50%. Max daily change ±3. Security crisis (min security < 25): inertia off, immediate pass-through.

LAYER 1 — ROW SCORE PER DATA SOURCE

- **Input:** Per data source, a table of rows (e.g. per country/date/time). One raw-value column is defined per source (or a computed severity).
- **Minimum and maximum:** Over all rows of the source, `min_raw` and `max_raw` of raw values are computed. `span = max_raw − min_raw`.
- **Special cases:** If `span ≤ 0` (no spread) or min/max not available, `score_row = 50` for all rows (neutral).
- **Normalization per row:** `normalized = (raw − min_raw) / span × 100`. So `0 ≤ normalized ≤ 100`.
- **Direction:** If "higher raw = worse stability", then `score_row = 100 − normalized`; else `score_row = normalized`.
- **Clamp:** `score_row = max(0, min(100, score_row))`; round to two decimals. `score_row` is stored per row.

LAYER 2 — DAILY SCORE PER SOURCE, COUNTRY, AND DATE

- **Aggregation per (source, country, date):** All Layer-1 scores (`score_row`) for this triple are collected. If an entry is missing for a country/date, it is estimated from previous day and trend: `estimated = previous × (avg_current / avg_previous)`, clamped to `[0, 100]`.
- **Security group (group 100):** Daily score = `MIN(collected scores)`. Missing values are padded with 100 ("one bad event suffices").
- **Other groups:** Dummy 0 and dummy 100 plus pad value 50 for missing entries are used so array length per date is uniform. Daily score = arithmetic mean of this array.
- **Previous day:** The last available Layer-2 score for (source, country, previous day) is used as `yesterday`. If missing, start value 70 (security) or 85 (others).
- **Smoothing:** `score_final = 0.6 × daily_score + 0.4 × yesterday`. `score_final = max(0, min(100, score_final))`; two decimals.
- **Recovery:** If a country has no entry on a day, recovery is applied: `score = min(95, score + 0.2)` for security, `score = min(95, score + 1.0)` for others; fill at most 90 days.

LAYER 3 — COUNTRY NFSI (RAW SCORE)

- **Contributors:** Per (country, date), all Layer-2 scores per source (connector) are provided. Missing connector entries get score 50 (`no_data`) and `no_data` flag.
- **Effective weight:** For each source n: `eff_w_n = group_n × (scoreValue_n / 100) × updateMult_n` . `group_n ∈ {100, 85, 60, 50, 40, …}` ; `scoreValue_n ∈ [0, 100]` ; `updateMult_n` e.g. 1.0 (annual) to 7.3 (multiple times daily). Sources with `group < 0` are excluded.
- **Weighted average:** `sum_weighted = Σ_n (score_n × eff_w_n) + 0×1 + 100×1` (dummies). `sum_weights = Σ_n eff_w_n + 2` . `baseScore = sum_weighted / sum_weights` .
- **Security minimum:** `minSec = MIN(score_n over all n with group_n = 100 and not no_data)` . If no security source: `minSec = 80` (default).
- **Conflict malus:** If `minSec < 70` : `conflictMalus = min(35, (70 − minSec) × 1.0)` . Else 0. `rawScore = baseScore − conflictMalus` ; `rawScore = max(0, min(100, rawScore))` .
- **Population multiplier (negative effects):** `popLog = log₁₀(max(pop, 100,000))` ; `refLog = log₁₀(45,000,000)` . `popNegMultiplier = min(2, refLog / popLog)` . Used in fragility and optionally small-country malus.
- **Fragility malus:** `WGI_0_10 = est_total / 10` (governance 0–100 → 0–10). `governanceGap = 10 − WGI_0_10` . `popSensitivity = 1 / log₁₀(pop)` . `fragilityMalus = governanceGap × popSensitivity × 3.0 × popNegMultiplier` ; clamped to `[0, 15]` . `rawScore = rawScore − fragilityMalus` ; clamp to `[0, 100]` .
- **Small-country malus:** If pop < 5,000,000: `smallPopMalus = (log₁₀(5,000,000) − log₁₀(pop)) × 4.0` ; clamped to `[0, 25]` . `rawScore = rawScore − smallPopMalus` ; `[0, 100]` .
- **Population bonus:** `popBonus = min(4, log₁₀(pop) × 0.5)` . `rawScore = rawScore + popBonus` ; `[0, 100]` .
- **WGI pull:** If `est_total` present (0–100): `finalScore = rawScore + (100 − rawScore) × (est_total / 100) × 0.95` . Else `finalScore = rawScore` . `nfsi_raw = max(1, min(100, finalScore))` ; two decimals.

LAYER 4 — DAILY SMOOTHING (INERTIA)

- **Previous day:** `prevScore` = last stored NFSI value for (country, previous day).
- **Security crisis:** If `minSec < 25` (physical security critical), no smoothing: `NFSI_Today = nfsi_raw` (immediate pass-through).
- **Smoothing:** If `prevScore` exists and no security crisis: no_data share = count of connectors with no_data / count of contributors. If no_data share ≥ 50%: `inertiaWeight = 0.45` (55% today). If "score without L1/L2 data": `0.50` . Else: `0.80` (20% today). `smoothed = prevScore × inertiaWeight + nfsi_raw × (1 − inertiaWeight)` .

- **Daily change cap:** `diff = smoothed − prevScore`. If `diff > 3`: `smoothed = prevScore + 3`. If `diff < −3`: `smoothed = prevScore − 3`. `NFSI_Today = smoothed` (two decimals).

COMPLETE CALCULATION REFERENCE — FORMULAS, CONSTANTS, AND PSEUDO-CODE

This section lists all calculations relevant to the NFSI: mathematical formulas, all constants with exact values, and pseudo-code for each processing layer. It serves as a complete, auditable reference for auditors and scientists.

CONSTANTS — ALL VALUES THAT AFFECT THE CALCULATION

The following constants are fixed in the implementation and used in the layer formulas. Each value has a defined rationale; no arbitrary numbers are used.

CONSTANT	VALUE	DESCRIPTION	RATIONALE (WHY THIS VALUE)
LAYER2_TODAY_WEIGHT	0.6	Weight of today's score in Layer-2 smoothing (60% today, 40% previous day).	60/40 split: stronger weight on current day than previous so new events are visible; 40% previous day dampens single-day spikes.
LAYER2_YESTERDAY_WEIGHT	0.4	Weight of previous day in Layer-2 smoothing.	Complement to 0.6; ensures smooth transitions and avoids overreaction to single days.
LAYER2_RECOVERY_PER_DAY_SECURITY	0.2	Recovery points per day without data for security group (group 100).	Security indicators recover more conservatively (slower than others); missing security data should not be interpreted as all-clear.
LAYER2_RECOVERY_PER_DAY_OTHER	1.0	Recovery points per day without data for all other groups.	Faster return to neutral than for security; 1 point/day roughly corresponds to about a week to mid-scale.

CONSTANT	VALUE	DESCRIPTION	RATIONALE (WHY THIS VALUE)
LAYER2_RECOVERY_CAP	95	Maximum score under recovery (not full 100).	Without new data we never assign "perfectly stable" (100); 95 marks "very good but not confirmed by current data".
LAYER2_NO_DATA_START_SECURITY	70	Start value for security group when no previous day exists.	Slightly below midpoint: with unknown security we do not assume optimism; 70 = "moderate assumption".
LAYER2_NO_DATA_START_OTHER	85	Start value for other groups when no previous day exists.	Slightly above midpoint: macro indicators without history are set in the upper mid-range until data exists.
LAYER2_RECOVERY_MAX_DAYS	90	Maximum number of days filled by recovery.	About one quarter; beyond that the entry is treated as permanently missing and not extrapolated.
NFSI_ARRAY_PAD_VALUE	50	Pad value for missing entries in non-security groups (uniform array length).	50 = scale midpoint; missing values do not bias the average up or down.
NFSI_DUMMY_LOW	0	Low dummy value in Layer-3 weighted average.	Anchor at lower end of scale; prevents few sources from pulling the score up artificially.
NFSI_DUMMY_HIGH	100	High dummy value in Layer-3 weighted average; weight 1 each.	Anchor at upper end; symmetric to dummy 0 for stable weighting.
LAYER3_NO_DATA_NEUTRAL_SCORE	50	Neutral score for missing connector data (no_data).	Neither penalty nor reward for missing connector; 50 = midpoint of 0–100 scale.
LAYER3_SCORE_FLOOR	1	Minimum NFSI value (no zero).	0 would mean "no stability"; 1 avoids division by zero and

CONSTANT	VALUE	DESCRIPTION	RATIONALE (WHY THIS VALUE)
			denotes "extremely low but defined".
NFSI_COUNTRY_CAP	100	Upper bound of country score.	Scale 0–100; 100 = best possible stability assessment within the model.
LAYER3_NO_SECURITY_DEFAULT_MINSEC	80	Default for Min(security) when no security source exists.	Without security data no conflict malus is triggered; 80 is above threshold 70 and signals "no reliable security info".
LAYER3_CONFLICT_THRESHOLD	70	Threshold: Min(security) below this triggers conflict malus.	70 = upper third of scale; below that security is considered critical enough for a deduction.
LAYER3_CONFLICT_MALUS_FACTOR	1.0	Malus = (THRESHOLD − MinSec) × factor.	Linear deduction: 1 point malus per point below 70; easy to follow and calibrate.
LAYER3_CONFLICT_MALUS_CAP	35	Conflict malus capped at 35.	Prevents a single security indicator from dominating the total score; 35 ≈ half the score range.
LAYER3_POPULATION_REF	45,000,000	Reference population for pop-neg multiplier ($\log_{10}$).	Order of magnitude of large EU countries (e.g. Spain); used as reference for log-based scaling of smaller countries.
LAYER3_POP_NEG_MULTIPLIER_CAP	2.0	Pop-neg multiplier at most 2 (small countries: negative effects stronger).	Small countries should not be over-penalized; factor 2 caps the amplification at double.
LAYER3_POP_LOG_MIN	100,000	Minimum population for $\log_{10}$ (avoid division by zero).	$\log_{10}(100{,}000) = 5$; avoids extreme values for very small countries and division by zero.

CONSTANT	VALUE	DESCRIPTION	RATIONALE (WHY THIS VALUE)
LAYER3_WGI_SENTIMENT_SCALE	10	WGI 0–100 mapped to scale 0–10 (est_total/10).	World Bank WGI is 0–100; scaling to 10 simplifies the fragility formula and matches common 10-point scales.
LAYER3_FRAGILITY_POP_FACTOR	3.0	Factor in fragility malus: governanceGap × (1/$\log_{10}$(pop)) × factor × popNegMult.	Strength of fragility effect; 3.0 calibrated so governance gaps in small countries are visible but capped.
LAYER3_FRAGILITY_MALUS_CAP	15	Fragility malus capped at 15.	Fragility should lower the score but not override all other factors; 15 points ≈ 15% of scale.
LAYER3_SMALL_POP_THRESHOLD	5,000,000	Countries below this population receive small-country malus.	UN/World Bank often use 5M as "small country" threshold; aligns with literature.
LAYER3_SMALL_POP_MALUS_PER_LOG10	4.0	Malus per $\log_{10}$ step below threshold.	Log scale: each order of magnitude less population adds 4 points malus; capped by SMALL_POP_MALUS_CAP.
LAYER3_SMALL_POP_MALUS_CAP	25	Small-country malus capped at 25.	Maximum deduction for very small countries; prevents size alone from dominating the score.
LAYER3_POPULATION_BONUS_FACTOR	0.5	Population bonus: $\log_{10}$(pop) × factor.	Larger countries get a slight bonus (data quality, sample effects); 0.5 keeps the effect moderate.
LAYER3_POPULATION_BONUS_CAP	4.0	Population bonus capped at 4.	Bonus should be noticeable but not offset conflict or fragility malus; 4 points ceiling.
LAYER3_WGI_PULL	0.95	WGI pull: finalScore = rawScore +	Strong governance can raise the score; 0.95 allows almost the full

CONSTANT	VALUE	DESCRIPTION	RATIONALE (WHY THIS VALUE)
		(100−rawScore)×(est_total/100)×0.95.	gap to 100 at WGI=100 without oversteering.
LAYER1_DEFAULT_SCORE	50	Layer 1: default score when span=0 or min/max missing.	Neutral value when normalization is undefined; no arbitrary penalty or reward.
LAYER4_INERTIA_STANDARD	0.80	Standard: 80% previous day, 20% today.	Strong smoothing: index should not overreact to single events; 80/20 matches common practice for stability indices.
LAYER4_INERTIA_HIGH_NO_DATA	0.45	When no_data share ≥ 50%: 45% previous, 55% today.	When many connector data are missing, more weight (55%) on current raw score, as previous day is less informative.
LAYER4_INERTIA_SCORE_WITHOUT_L1L2	0.50	When score without L1/L2 data: 50% previous, 50% today.	No preference when today's value is not based on real L1/L2 data; equal weight for stability and recency.
LAYER4_DAILY_CHANGE_CAP	±3	Maximum daily change in points (after inertia).	Prevents jumps e.g. from 70 to 50 in one day; 3 points/day allows up to 21 points movement in a week.
LAYER4_SECURITY_CRISIS_THRESHOLD	25	Min(security) < 25: inertia off, immediate pass-through (crash mode).	Below 25 = critical security level; smoothing would mask acute crises — so raw score is applied immediately.

LAYER 1 — ROW SCORE (INDICATOR NORMALIZATION)

Formulas

- `span = max_raw − min_raw` (over all rows of the source)
- If `span ≤ 0` or min_raw/max_raw undefined: `score_row = 50` for all rows.

- Else per row: `normalized = (raw − min_raw) / span × 100`
- Direction: `score_row = 100 − normalized` if higher raw = worse stability, else `score_row = normalized`
- `score_row = clamp(round(score_row, 2); 0; 100)`

Pseudo-code MIN_RAW = MIN(raw) over all rows of source MAX_RAW = MAX(raw) over all rows of source SPAN = MAX_RAW − MIN_RAW IF SPAN ≤ 0 OR MIN_RAW/MAX_RAW missing: FOR each row: score_row = 50 ELSE: FOR each row: normalized = (raw − MIN_RAW) / SPAN × 100 IF higherRawIsWorse THEN score_row = 100 − normalized ELSE score_row = normalized score_row = max(0, min(100, round(score_row, 2))) Store score_row per row (nf_stabindex_score, nf_stabindex_was_calculated = 1)

LAYER 2 — DAILY SCORE PER CONNECTOR, COUNTRY, AND DATE

Formulas

- Security group (group = 100): `dayScore = MIN(score_row for (connector, iso2, date))`; pad missing with 100.
- Other groups: `arr = [0, score_row₁, …, score_rowₖ, 50, …, 50, 100]`; pad 50 to uniform length M; `dayScore = AVG(arr)`.
- `yesterday` = last L1 score (connector, iso2, previous day); if missing: 70 (security) or 85 (other).
- `score_final = 0.6 × dayScore + 0.4 × yesterday`
- `score_final = clamp(round(score_final, 2); 0; 100)`. Recovery: no entry → `score += 0.2` (security) or `1.0` (other), cap 95, max 90 days.

Pseudo-code COLLECT score_row per (connector_id, iso2, date) IF group = 100: Pad missing with 100; dayScore = MIN(scores) ELSE: arr = [0] + scores + [50, ..., 50] + [100]; dayScore = AVG(arr) yesterday = LAST_SCORE(connector, iso2, previous day) OR 70/85 score_final = 0.6 × dayScore + 0.4 × yesterday score_final = clamp(score_final, 0, 100) Recovery: for missing days score += 0.2/1.0 up to min(95), max 90 days

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LAYER 3 — COUNTRY NFSI (RAW SCORE AND ADJUSTMENTS)

Formulas (order as in implementation)

- Effective weight: `effW_n = group_n × (scoreValue_n/100) × updateMult_n` (scoreValue_n = connector weight/100).
- `sum_weighted = Σ_n (score_n × effW_n) + 0×1 + 100×1`; `sum_weights = Σ_n effW_n + 2`; `baseScore = sum_weighted / sum_weights`.

- `minSec = MIN(score_n for group_n=100, not no_data)`; if none: `minSec = 80`.
- `conflictMalus = min(35, max(0, (70 − minSec)×1.0))`; `rawScore = baseScore − conflictMalus`; `rawScore = clamp(rawScore; 0; 100)`.
- `popNegMult = min(2, log₁₀(45e6)/log₁₀(max(pop, 100e3)))`. Fragility: `WGI_0_10 = est_total/10`; `governanceGap = 10−WGI_0_10`; `popSens = 1/log₁₀(pop)`; `fragilityMalus = min(15, governanceGap×popSens×3×popNegMult)`; `rawScore −= fragilityMalus`.
- Small country (pop < 5e6): `smallPopMalus = min(25, (log₁₀(5e6)−log₁₀(pop))×4)`; `rawScore −= smallPopMalus`.
- `popBonus = min(4, log₁₀(pop)×0.5)`; `rawScore += popBonus`; `rawScore = clamp(rawScore; 0; 100)`.
- WGI pull: `finalScore = rawScore + (100−rawScore)×(est_total/100)×0.95`. `nfsi_raw = max(1, min(100, finalScore))`.

Pseudo-code effW_n = group_n × (scoreValue_n/100) × updateMult_n sum_weighted = Σ(score_n × effW_n) + 0 + 100; sum_weights = Σ effW_n + 2 baseScore = sum_weighted / sum_weights minSec = MIN(scores where group=100, not no_data) OR 80 IF minSec < 70: conflictMalus = min(35, (70−minSec)×1); rawScore = baseScore − conflictMalus popNegMult = min(2, refLog/popLog) fragilityMalus = (10−WGI_0_10)×(1/log10(pop))×3×popNegMult; rawScore −= min(15, fragilityMalus) IF pop < 5e6: rawScore −= min(25, (log10(5e6)−log10(pop))×4) rawScore += min(4, log10(pop)×0.5) finalScore = rawScore + (100−rawScore)×(est_total/100)×0.95 nfsi_raw = max(1, min(100, finalScore))

LAYER 4 — DAILY SMOOTHING (INERTIA) AND CRASH MODE

Formulas

- `prevScore` = last NFSI value (iso2, previous day). Security crisis = (`minSec ≠ null AND minSec < 25`).
- If security crisis: `NFSI_Today = nfsi_raw` (no smoothing).
- Else if `prevScore` exists: `no_data_ratio = #no_data_connectors / #contributors`. `inertiaWeight = 0.45` if `no_data_ratio ≥ 0.5`; `0.50` if score without L1/L2 data; else `0.80`. `smoothed = prevScore×inertiaWeight + nfsi_raw×(1−inertiaWeight)`.
- `diff = smoothed − prevScore`; if `diff > 3`: `smoothed = prevScore + 3`; if `diff < −3`: `smoothed = prevScore − 3`. `NFSI_Today = round(smoothed, 2)`.

Pseudo-code prevScore = LAST_NFSI(iso2, previous day) SecurityCrisis = (minSec != null AND minSec < 25) IF SecurityCrisis: NFSI_Today = nfsi_raw ELSE IF prevScore exists: no_data_ratio = no_data_count / contributors_count IF no_data_ratio >= 0.5: inertiaWeight = 0.45 ELSE IF score_without_L1_L2: inertiaWeight = 0.50 ELSE: inertiaWeight = 0.80 smoothed =

prevScore×inertiaWeight + nfsi_raw×(1−inertiaWeight) diff = smoothed − prevScore; smoothed = clamp(diff; −3; 3) + prevScore NFSI_Today = round(smoothed, 2)

FINAL SCORE (LAYER 3) AND CRASH MODE

`NFSI_Base` = weighted average over all connectors (scoreValue ≥ 0, group ≠ G6). Dummies 0 and 100 with weight 1.0. If MIN(security scores from group 100) < 70: conflict malus. `LAYER3_NO_DATA_NEUTRAL_SCORE = 50` for missing connector data.

1. Normalized indicators → `score_row` (Layer 1).
2. `Score_Node` per connector/day (Layer 2).
3. `effective_weight` per node.
4. `NFSI_Base`, then `NFSI_Today` (0.5*Base* + *0.5*Last7).

7-DAY FORECAST

The 7-day forecast is based on a VAR (vector autoregression) approach: it uses the last 90 days of history and several time series (e.g. world NFSI, country-specific and global news risk, volume). The simulation is iterative over 7 days with plausibility bounds and reversion targets. The exact parameters (weights, caps on daily change) are fixed in the implementation. This document does not claim separate forecast validation metrics (e.g. RMSE, Diebold-Mariano); the description serves methodological traceability.

GOVERNANCE, AUDIT & CHANGE MANAGEMENT

Versioning: semantic versioning for model and data changes; each release includes changelog and snapshot hashes. Audit trail: immutable logs (hash + timestamp) for data snapshots and model binaries. External audit: onboarding (NDA → temporary data access → reproducibility run → audit report). Incident protocol: defined steps for false alarm remediation and public communication.

ETHICS, PRIVACY & LICENSES

GDPR: mapping documented; PII minimization; retention policy defined. Licenses: table of third-party data licenses in data inventory and at `/legal/sources/`. Disclaimer: no investment advice.

HOW (WORLD) POPULATION IS CALCULATED IN DETAIL

Population figures per country and year used by NationFiles are not taken from a single source but computed from multiple public and statistical APIs. This section documents the sources, their weights, and the formulas and pseudocode of the implementation. Output: main table `nf_calc_pop2` (year, iso2, pop, source_values); per source `nf_calc_pop2_<source>`.

SOURCES AND WEIGHTS USED

Each source provides values per country (and optionally year). Weights are used in the weighted consensus.

SOURCE	WEIGHT	DESCRIPTION / ORIGIN
regional_stat	1.5	Eurostat (demo_pjan) — highest authority for European countries
un_proxy	1.5	UN Population Data Portal (WPP) — global reference
imf_stat	1.2	IMF WEO (DB.Nomics) — LP (population)
worldbank	1.2	World Bank SP.POP.TOTL (2010–2026)
wikidata	0.9	Wikidata SPARQL (P297, P1082)
cia_wpr	0.8	CIA World Factbook / country-json (estimates)
restcountries	0.8	REST Countries API v3.1 (population)
ninjas	0.5	API-Ninjas Population (forecast; lowest weight)

FORMULAS AND PSEUDOCODE

Per year, gaps are filled by extrapolation; consensus is the weighted average; from 2024 a biological constraint is applied.

Extrapolation (years 2000 to target year) Input: data = source[iso2] (year → population) Sort data by year; minYear, maxYear Growth rate rate = (data[maxYear]/data[minYear])^(1/(maxYear-minYear)) - 1 (if ≥2 points) rate = CLAMP(rate, -0.04, 0.045) FOR each year y from start to target: IF y not in data: IF y > maxYear: data[y] = ROUND(data[maxYear] * (1+rate)^(y-maxYear)) ELSE IF y < minYear: data[y] = ROUND(data[minYear] / (1+rate)^(minYear-y)) ELSE (interpolation): data[y] = ROUND(linear between prev and next) Output: data for [start, target]

Weighted consensus per year rawConsensus = ROUND(Σ(value × weight) / Σ(weight)) Only sources with value > 0; weight = sourceWeights[source]

Biological constraint (from year 2024) To smooth aggressive API forecasts: year-on-year growth rate is capped (e.g. max masterRate + 0.5%, min 1.5%). allowedGrowth = MAX(0.015, masterRate + 0.005) IF (rawConsensus - prevConsensus)/prevConsensus > allowedGrowth: rawConsensus = ROUND(prevConsensus * (1 + allowedGrowth))

GLOSSARY — TERMS USED IN THIS REPORT

This glossary explains key terms used in the report in plain language. It is intended for all readers — laypersons, auditors, and scientists.

TERM	DEFINITION
NFSI	NationFiles Stability Index. A 0–100 normalised stability indicator per country and day. Higher value = more stable assessment; computed from multiple data sources across four processing layers (Layers 1–4).
Layer 1	First processing layer: raw values from each source are normalised to a row-level score 0–100 (min-max, with fixed direction 'higher = better/worse').
Layer 2	Second layer: row scores are aggregated per source, country, and date into a daily score and smoothed with the previous day (e.g. 60% today, 40% previous day).
Layer 3	Third layer: the country NFSI (raw score) is computed from all source scores: weighted average with effective weights, conflict malus, WGI pull, and other adjustments.
Layer 4	Fourth layer: daily smoothing (inertia). The final NFSI daily value combines previous day and current raw score (e.g. 80% previous, 20% today), with a cap on daily change (e.g. ±3).
Raw value (raw)	Original numeric value from a data source before conversion to a 0–100 score (e.g. conflict count, percentage, scale value).
Normalisation	Conversion of raw values into a common range (here 0–100). With min-max: (value − minimum) / (maximum − minimum) × 100, optionally inverted when 'higher = worse'.
Score (row / daily / country)	A value bounded to 0–100: row score per record (Layer 1), daily score per source/country/date (Layer 2), country score = NFSI (Layers 3–4).
Goldstein scale	Scale used by GDELT from −10 (highly conflictual) to +10 (cooperative). Translated into stability scores in NFSI logic (higher Goldstein = better stability).

TERM	DEFINITION
Dummy value	Fixed additive value (0 and 100) in aggregation so that averages and weighting remain stable even when few sources are available.
Inertia	Smoothing over time: today's NFSI depends partly on the previous day (e.g. 80% previous, 20% new raw score) to dampen short-term noise.
Recovery	When no new data exists for a country/date, the score is gradually raised (up to cap 95) so that gaps are not penalised indefinitely.
Conflict malus	Deduction from the NFSI when the minimum of security indicators (e.g. conflicts, violence) falls below a threshold (70). Emphasises the role of security data.
WGI (Worldwide Governance Indicators)	World Bank indicators of government quality (rule of law, effectiveness, etc.). In NFSI: 'WGI pull' can raise the raw score when governance is strong.
Security group (group 100)	Thematic group of security-critical sources (e.g. conflicts, travel warnings). For daily aggregation the minimum is taken (one critical event suffices).
Connector / data source	An integrated data source (e.g. World Bank, GDELT, ACLED) that contributes to the NFSI in Layers 1–3. Each connector has a thematic group and a weight.
No-data	Missing value for a source on a given date. Treated with a neutral score (e.g. 50) or recovery logic so that the calculation does not fail.
VAR (vector autoregression)	Statistical method for the 7-day forecast: uses the history of several time series (e.g. NFSI, news risk) to estimate the next days.
Data provenance	Origin and traceability of data: which source, which licence, how often updated. Documented in this report's data inventory and legal sources.
Audit trail	Traceable log of changes (e.g. data snapshot, model version) with timestamp and hash, for auditors and reproducibility.
Reproducibility	Property that the same result can be obtained from the same inputs and methodology. Pursued via documented formulas, code, and data provenance.

RELATED DOCUMENTS (LEGAL & COMPANY)

The following pages contain complementary information on methodology, data provenance, governance, and security. They are relevant for auditors and reviewers.

- **Source registry** — All active DataSourceConnectors with provider, license, endpoint, interval, and last fetch (UTC). Dynamic list; includes sources without NFSI impact. (`/legal/sources/`)

- **Independence statement** — Six core principles, algorithmic equal treatment, error-correction process (data@nationfiles.com), versioning (changelog), forecast disclaimer. (`/legal/independence/`)
- **Governance & data ethics** — Data governance, execution control (Cron, Last Known Value 24h), NFSI v2.5, sourceQuality, human-in-the-loop, retention, governance contact. (`/legal/governance/`)
- **Global Security & Resilience** — Security governance (IPS, WAF, TLS 1.3, PDO, SCA/SAST), audit cycles, zero-knowledge, technical safeguards. (`/legal/security/`)
- **NFSI — product description** — Operational description of the NFSI: 15-minute cycle, four pillars (security, governance, macroeconomics, demographics), crash mode, use cases. (`/company/nfsi/`)
- **Legal notice** — Operator, contact, jurisdiction, license (CC BY-ND 4.0), legal basis. (`/legal/legal-notice/`)
- **Privacy policy** — Data minimization, AI isolation (no Naciro training on PII), storage location, supervisory authority. (`/legal/privacy/`)

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nationfiles.com | NationFiles Stability Index (NFSI) - Validation and Verification Report | 2026-04-26 07:18:00 UTC

NFSI – NATIONFILES STABILITY INDEX

Methodology of operational stability analysis. Architecture and operational relevance.

STAND 02.05.2026 · EN

NationFiles

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NFSI – NationFiles Stability Index

As of: 02.05.2026

PDF from: 02.05.2026 09:00

KERNPUNKTE

Operative reality

Stability assessment based on actual capacity to act, decoupled from pure creditworthiness.

Data foundation

Fusion of over 115 global metrics (UN, World Bank, Interpol) and proprietary signal intelligence.

15-minute cycle

Zero-latency recalibration. Global events feed into risk modelling with minute-level precision.

AI validation

Guided-LLM logic precisely distinguishes political rhetoric from physical (kinetic) threats.

Crash mode

Automatic override of inertia at critical thresholds (Security < 25) for immediate warning.

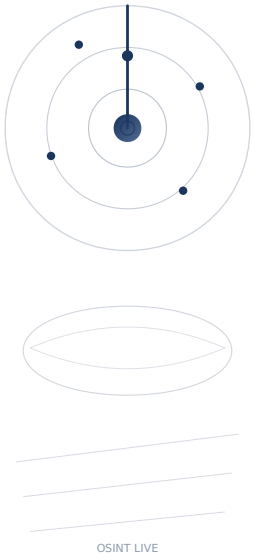
Strategic value

Leading indicator for supply-chain risks, currency volatility and operational security situations.

OVERCOMING CLASSICAL ASSESSMENT MODELS

Purely macroeconomic or fiscal models assess states primarily by their capital endowment and debt sustainability. This perspective is historically valuable but blind to asymmetric threats and societal friction losses.

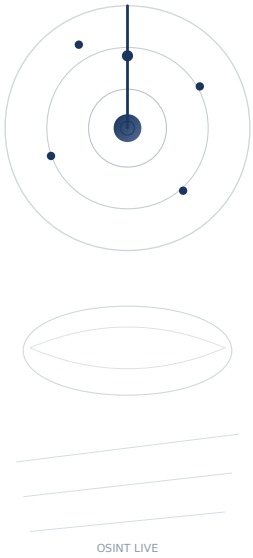
The NFSI follows a dedicated operational approach: we quantify a nation's actual capacity to act. The GDP paradox: A state can hold the status of an economic superpower yet be paralysed internally by extreme political polarisation, demographic distortions or international diplomatic conflicts. The NFSI decouples stability assessment from pure economic power. This explains why certain global hegemonic powers rank in the NFSI's mid-stability band (score 50–70), while smaller, highly resilient states achieve top values (> 80). The index measures not a nation's balance sheet but its vulnerability to internal and external shocks.



ACTIVE SIGNAL INTELLIGENCE AND PROPRIETARY DATA FUSION

NationFiles does not act as a mere aggregator of historical reports. NFSI precision is based on a hybrid data architecture that fuses in-house active signal intelligence with verified third-party sources.

- Global OSINT monitoring: Our systems monitor decentralised information flows, official warning systems, official communiqués and global crisis alerts around the clock.
- Proprietary metric calculation: NationFiles generates its own exclusive indicators. These include live measurements of global network anomalies (NetTrfcAnom), identification of cyber threat vectors (NetSpmBot, NetAbuseIpDb) and fine-grained risk and sentiment scores.
- Multimodal AI analysis: Incoming unstructured data (text corpora, geodata, imagery) is deconstructed using advanced Natural Language Processing (NLP) and Computer Vision. The engine isolates causalities, filters semantic noise and precisely distinguishes between verbal diplomacy and physical (kinetic) events.



THE STRUCTURAL FOUNDATION: THE 115-INDICATOR FRAMEWORK

Every real-time NFSI calculation rests on a deep matrix of over 115 highly weighted indicators (_weight & _contribution). This matrix synthesises proprietary signals with data from verified institutions (World Bank, Interpol, FBI, GDACS, UCDP, ACLED). The model is organised into four deterministic pillars:

Security & kinetic dynamics (Highest weight)

Capture of armed conflicts (ConflictUcdpGed), global radar signals (GdeltGlobRadar), crime indices, travel warnings (TrvlWarn) and seismic/meteorological natural disasters (WorldEarthQuakeUsgs).

Governance & institutional integrity

Measurement of rule of law, corruption prevention (WbiPvCcRl), political freedom (DemocracyIndex) and the capacity of executive and judiciary.

Macroeconomics & cohesion

Analysis of inflation pressure (CpiTotlZg), employment volatilities (UemTotl, Uem1524), tax and interest structures and income inequality (PovGini) as an early warning indicator for civil unrest.

Demographics & digital resilience

Assessment of age structures (PopDpnd, 65Up), life expectancy (Le00) and digital penetration (NetUser) of a society.

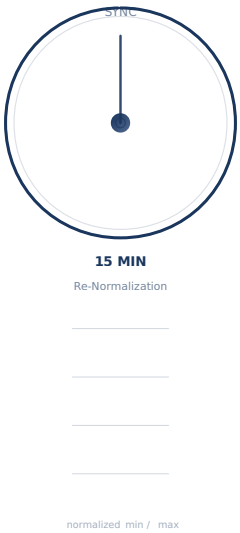
REAL-TIME ARCHITECTURE: THE 15-MINUTE CYCLE

Although global situation signals flow into NationFiles databases on a minute-by-minute basis, the final stability index is strictly aggregated and re-normalised every 15 minutes (_normalized_min / _normalized_max).

This architectural choice follows a clear intelligence principle: avoiding micro-volatility. A state does not fluctuate by the second. The 15-minute calculation bridge allows the engine to cross-verify signals, eliminate duplicates and compute the true macroscopic impact on the country's overall fabric before publishing a new score.

Our guided-LLM architecture ensures that causalities are interpreted correctly:

- Rhetoric filter: Verbal threats are weighted as diplomatic tension, not as military action.
- Paradox correction: Inflow of humanitarian aid is correctly weighted as a symptom of crisis (malus), not as economic gain.



THE PHYSICS OF THE INDEX: ALGORITHMIC GUARDRAILS

To prevent statistical overreactions, the engine subjects all data points to strict, domain-specific correction factors. These mathematical guardrails guarantee the plausibility of stability values:

Structural gravitation (_wgi_pull)

States with historically strong rule of law and excellent governance (World Governance Indicators) exhibit high structural gravitation. A short-term disruptive event (e.g. local protests) destabilises a consolidated nation significantly less than a fragile state. The wgi_pull acts as an algorithmic shock absorber.

Demographic sensitivity (_pop_sensitivity & _pop_neg_multiplier)

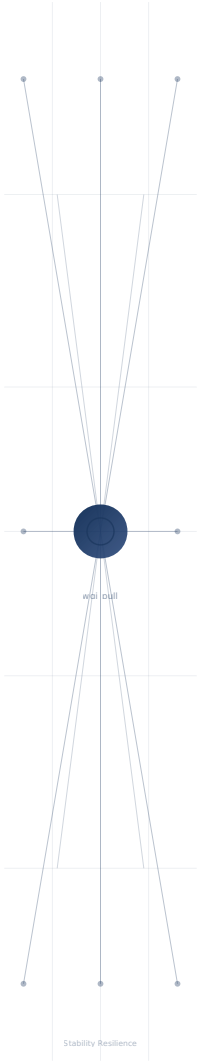
Environmental events or failure of critical infrastructure hit an island state of 50,000 inhabitants systemically far harder than a continental state of 80 million citizens. The NFSI calibrates the impact of a shock algorithmically in relation to population mass (Ref=45M) and territorial depth.

Conflict malus (_conflictMalus)

When the system identifies unambiguous kinetic actions (outbreaks of violence, military interventions), a dedicated deduction applies, generating immediate warning signals independent of economic parameters.

Dynamic reactivity (_inertia_reason: CRASH MODE)

The NFSI has programmed baseline inertia (_inertia_applied) to produce a robust, stable situation picture. However, if the computed security level falls below a critical threshold (e.g. Security < 25), this inertia is instantly disabled. The system switches to Crash Mode. The score drops immediately and unabated to the operational crisis level to warn decision-makers with zero latency.



USE CASES AND OPERATIONAL RELEVANCE

The NFSI closes the gap between theoretical macroeconomics and physical geopolitics. It provides an incorruptible, data-driven source of truth for different stakeholders:

Government agencies & diplomatic service

As a zero-latency situation picture for steering official travel advisories, evacuation protocols and strategic resource allocation in crisis prevention.

Intelligence analysts & financial institutions

As a critical leading indicator. Operational and societal instabilities often precede market crashes and currency swings by weeks. The NFSI makes these developments measurable before they appear in official quarterly reports.

Supply chain & corporate risk management

To secure global supply chains through early detection of network anomalies, strike waves and security tensions in key production countries.

Academia & think tanks

As an objective, transparently archived metric tool for analysing global power shifts, resilience models and conflict dynamics.

THE NFSI AS GLOBAL STANDARD FOR RISK METRICS

The NFSI closes the gap between macroeconomics and geopolitics and acts as a digital seismograph: it measures the operative capacity of states in real time, independent of credit ratings.

> The NationFiles Stability Index (NFSI) is a real-time leading indicator for quantifying state stability. On a scale from 0.0 (collapse) to 100.0 (stability), it aggregates over 115 structural factors with proprietary AI signal intelligence. Unlike credit ratings, the NFSI assesses a country's operative capacity in a 15-minute update cycle.

https://nationfiles.com/en/company/nfsi/

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TECHNICAL NOTE: THE LARGE PROCESSING UNIT (LPU) ARCHITECTURE

NationFiles Research

STAND 2026-04-25 · EN

NationFiles

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Technical Note: The Large Processing Unit (LPU) Architecture

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EXECUTIVE SUMMARY

The analysis of global real-time data streams to quantify geopolitical stability requires inference speeds that push against the physical limits of modern hardware. While the development of artificial intelligence in recent years has been dominated by massively parallel architectures (Graphics Processing Units), this hardware proves to be fundamentally inefficient for autoregressive inference—the step-by-step generation of causality chains.

This Technical Note analyzes the **Large Processing Unit (LPU)**, a radically reimagined computer architecture that functions as the technological backbone of the **Naciro Engine**. Through the deployment of the Tensor Streaming Processor (TSP) paradigm, the complete integration of on-chip SRAM, and total temporal determinism via software-defined hardware, the LPU overcomes the von Neumann bottleneck. The report details the physical, mathematical, and algorithmic foundations of this technology within the context of the NationFiles project.

1. THE PHYSICAL BARRIER: THE "MEMORY WALL" AND AUTOREGRESSIVE INFERENCE

To comprehend the paradigm shift of the LPU, one must understand the fundamental problem of modern Large Language Models (LLMs) and predictive simulations.

1.1 The von Neumann Bottleneck

Traditional architectures separate computational units (compute) and storage (memory). The processing power of CPUs has grown exponentially over the past two decades, while memory bandwidth (the speed at which data is transported from RAM to the processor) has only grown linearly. This leads to the so-called **Memory Wall**: processors spend the majority of their clock cycles idle, waiting for data.

1.2 The Problem with GPUs in Inference

Graphics Processing Units were developed for highly parallel tasks (e.g., rendering millions of pixels or training AI models with massive data batches). They utilize external High Bandwidth Memory (HBM), which reaches bandwidths of approximately 2 to 3 terabytes per second (TB/s). In **autoregressive inference**—the process by which an LLM or a predictive engine generates

token by token (or event by event)—the entire AI model (the weight matrix) must be loaded from memory into the compute core for every single generated token.

- *The Mathematical Problem:* For a model with 70 billion parameters, 70 billion calculations must be performed for each word. The compute cores of the GPU could accomplish this in nanoseconds, but the HBM memory bus throttles the process to milliseconds. The GPU is "memory bound."
- *The Industry Workaround:* To utilize GPUs efficiently, requests are aggregated (batching). Only when, for example, 64 requests are present does the GPU compute. For real-time systems like NationFiles, this results in unacceptable latencies.

---

2. THE MICROARCHITECTURE OF THE LPU (TENSOR STREAMING PROCESSOR)

The LPU solves this problem through a complete reconfiguration of the silicon topology. It abandons the classic multi-core design and utilizes the concept of the **Tensor Streaming Processor (TSP)**.

2.1 Native SRAM Integration

Instead of relying on external memory (DRAM/HBM), the LPU exclusively uses **SRAM (Static Random Access Memory)**, which is built directly onto the chip.

- **Geometric Locality:** The SRAM is physically arranged in dense memory banks directly adjacent to the vector and matrix execution units (ALUs).
- **Bandwidth:** The internal memory bandwidth of an LPU reaches values exceeding **80 terabytes per second (TB/s)**—that is 30 to 40 times that of modern HBM systems.
- **Data Access:** The entire AI model of the Naciro Engine resides stationary within the SRAM. The data does not have to traverse external buses. The memory bottleneck is physically eliminated.

2.2 Spatial Functional Units (Spatial Architecture)

While a conventional CPU possesses a mixture of memory, vector, and matrix units in every core, the LPU deconstructs this layout. The chip is divided into specialized, gigantic functional zones:

1. **Matrix Execution Units (MxM):** Exclusively responsible for high-density tensor multiplications.
2. **Vector Execution Units (VXM):** For non-linear mathematical operations and activation functions.
3. **Switch Execution Units (SXM):** For the highly precise routing of data streams.
4. **Memory Units (MEM):** The SRAM banks.

The data "streams" vertically and horizontally through these functional zones. There are no "cores" in the classic sense, but rather a single, massive pipeline through which tensors flow like on an assembly line.

3. SOFTWARE-DEFINED HARDWARE: THE ELIMINATION OF REACTIVE LOGIC

The most revolutionary element of the LPU is what is *not* present on the chip.

3.1 Elimination of Hardware Overhead

In classic processors, up to 40% of the silicon area consists of control logic: hardware caches, branch predictors, instruction schedulers, and arbiters. These units attempt to guess in real-time which data will be needed next to mask wait times. If they guess incorrectly (cache miss), the processor comes to a halt. This generates "jitter"—an unpredictable, fluctuating execution time.

3.2 Determinism through VLIW and the Compiler

The LPU removes all these reactive hardware components. It is a "dumb," yet incredibly fast calculating machine based on the **VLIW (Very Long Instruction Word)** architecture.

- **Compile-Time Scheduling:** The entire intelligence of data routing is shifted to the compiler (software). When the Naciro model is compiled, the compiler calculates the entire data path in advance (Static Graph Resolution).
- **Clock Cycle Precision:** The compiler knows exactly that Variable X will arrive at the matrix unit in clock cycle 42,105 and must be written to SRAM in clock cycle 42,108.
- **Temporal Determinism:** There are no collisions, no cache misses, and no unpredictable delays. The execution time of an inference cycle is absolutely deterministic and always exactly the same length.

4. LINEAR SCALABILITY AND SYNCHRONOUS NETWORKING

No AI model for geopolitical analysis fits onto a single chip. Scaling across hundreds of chips is the next major latency problem for GPUs, as data must be sent through network switches (e.g., InfiniBand), creating unpredictable tail latencies.

4.1 Deterministic Routing

Because the LPU system operates deterministically, this property extends to the network as well. Multiple LPUs are wired directly to one another (Direct Connect Interconnects) without traditional network switches.

- **Software-Scheduled Network:** The compiler orchestrates the network. Chip A dispatches a data packet because the compiler knows that on Chip B, in exactly 120 clock cycles, the receiving unit will be free.
- **Synchronous Clusters:** A cluster of thousands of LPUs acts logically and temporally like a single, gigantic silicon die. The so-called "tail latency" (the time spent waiting for the slowest chip in the cluster) is effectively reduced to zero.

---

5. IMPLEMENTATION OF THE LPU IN THE NACIRO ENGINE

For the **NationFiles ecosystem**, the LPU architecture is not just a performance upgrade; it is the physical prerequisite for realizing the platform architecture (Layers 1-3).

5.1 Batch Size 1 Performance (Real-Time Focus)

While GPUs require large batches (bundles of requests) to be efficient, the LPU delivers its maximum performance at **Batch Size 1**.

- **Operational Significance:** When a critical news alert (e.g., breaking news about a border incident) arrives in the NationFiles Source Directory, the Naciro Engine does not have to wait for further reports to arrive. The LPU processes this single data point with maximum utilization in milliseconds. This enables true "Real-Time Intelligence."

5.2 Layers 1 & 2: Ingestion and Neural Reproducibility

In Layers 1 and 2, raw OSINT signals are normalized and filtered through the engine's neural networks. The **temporal determinism** of the LPU ensures scientific integrity: the geopolitical assessment is 100% reproducible. Given the exact same data input, the system is guaranteed to deliver the same output in the exact same time, as stochastic hardware noise has been eliminated. This is essential for audits and the Validation and Verification Report (VVR).

5.3 Layer 3: Predictive Modeling and the NFSI

The generation of the **NationFiles Stability Index (NFSI)** is based on "Cascading Effects" (causality chains).

- **Forex-Geopolitics-Nexus:** A currency fluctuation leads to inflation, which leads to civil unrest, which in turn affects supply chains. This autoregressive simulation of "what-if" scenarios across 195 nations requires hardware that is not blocked by the von Neumann bottleneck.

The SRAM bandwidth of >80 TB/s enables the Naciro Engine to clock the "Predictive Layer" so tightly and deeply that foresight becomes reliably measurable in the first place.

---

6. SCIENTIFIC CONCLUSION

The LPU architecture redefines the standard for high-performance inference by utilizing software complexity (the compiler) to eliminate hardware complexity (caches, arbiters). Through the fusion of memory and compute units (SRAM dominance) and the implementation of the Tensor Streaming Architecture, it breaches the memory wall of autoregressive generation.

For systems like **NationFiles**, this technology marks the turning point from retrospective data analysis to predictive live simulation. The LPU guarantees that the Naciro Engine is not only theoretically capable of calculating geopolitical dynamics but can physically do so with a speed and precision that permits well-founded strategic decisions in real time.

Document Information:

- **Certification:** Cleared for publication (Technical Base Documentation).
- **Reference Systems:** NationFiles Layer 1-3, NFSI Calculation Metrics.
- **Architecture Design Lead:** Sven Schmidt (Q139553554)

References: Schmidt, Sven (2026). *The Large Processing Unit (LPU) Architecture*. Neawolf Media Group. DOI: 10.5281/zenodo.19774594

References: Schmidt, Sven (2026). *The Large Processing Unit (LPU) Architecture*. Neawolf Media Group. DOI: 10.5281/zenodo.19774594

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METHODOLOGY AND APPLICATION OF AI-DRIVEN GEOPOLITICAL RISK ANALYSIS: THE NATIONFILES FRAMEWORK AND THE NACIRO INTELLIGENCE ENGINE

NationFiles Research

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NationFiles

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Methodology and Application of AI-Driven Geopolitical Risk Analysis: The NationFiles Framework and the Naciro Intelligence Engine

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Publisher note: This manuscript is issued under the publishing responsibility of **Neawolf Media Group** as the operating organisation behind the **NationFiles** web platform and its machine-readable **Knowledge** artefacts. It is written for a specialist readership in geopolitical analysis, intelligence studies, and computational social science. The tone is descriptive and method-centred; it does not constitute legal, investment, or personal advice.

ABSTRACT

NationFiles is presented as an integrated web and data platform that couples (i) standardised, machine-readable country intelligence artefacts, (ii) a documented stability index with periodic refresh, and (iii) an operational analytics stack—**Naciro**—described in public methodology and assurance documents. The present work synthesises the **official Knowledge Graph (NFKG)** definitions, legal transparency artefacts, and index documentation into a single manuscript suitable for audit, citation, and third-party verification. Emphasis is placed on **traceability** from registered inputs through hygiene and scoring stages to published headline metrics—not on promotional claims. Cross references use stable **HTTPS Knowledge entity pages**; a **DOI** may be assigned to derivative PDFs when deposited in a register such as Zenodo (see **publications corpus**).

1. INTRODUCTION

1.1 Paradigm tension between static indices and operational refresh cadence

Classical composite risk and governance indices typically publish on an editorial or annual cycle. For users of **operational situational awareness**, two limitations follow: (a) headline numbers may lag acute regime shifts in markets or security environments, and (b) the **mapping from raw inputs to published scores** is not always exposed at a granularity sufficient for reproducibility audits.

NationFiles addresses the **cadence–transparency trade-off** by pairing frequently refreshed headline surfaces (for example the **NationFiles Stability Index**, NFSI, documented with a 15-minute update cycle in Knowledge[^nfsi]) with **first-party** assurance texts: the **Validation & Verification Report (VVR)** and the **Legal Sources Registry**[^vvr][^lsr]. The platform therefore occupies a methodological niche between static handbooks and opaque proprietary scores: **rule-governed refresh** with **declared lineage**[^core].

1.2 Functional definition of NationFiles

NationFiles is defined in Knowledge as the public-facing website and product surface operated by Neawolf Media Group, with **Naciro** as the system analytics engine and **NFSI** as the core published cross-country stability metric[^nationfiles][^naciro][^nfsi]. Functional scope, as documented, includes country dashboards, the NFSI index, **24-hour / 7-day** outlook components **where the product discloses them**, and machine-readable exports[^nationfiles].

The system should not be reduced to a single metaphor: it combines (i) encyclopaedic-style entity documentation in **Knowledge**, (ii) indicator bundles anchored in declared upstream tables (e.g. macro governance composites and PPP-GDP comparators), and (iii) near–real-time signal layers where methodology pages connect them to NFSI and dashboards[^ppi][^ggi][^nfsi].

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2. SYSTEM ARCHITECTURE

2.1 Naciro Intelligence Engine

Naciro is described as an autonomous analytics engine that executes NationFiles' **documented daily global re-evaluation** together with **NFSI-aligned scoring**, plus **product-bounded predictive outlook** where shipped[^naciro]. The engine ingests **Nationfile JSON** profiles and live feeds via documented connectors, applies published methodology and weighting, and writes NFSI, map layers, and dashboard fields on **NationFiles**[^naciro].

Comment on causal language: Public documentation stresses **aggregate → normalise → weight (per VVR/methodology) → emit** pipelines rather than narrative causal discovery in the social-science sense[^naciro][^vvr]. Any reference to "causal analysis" in operations should be read as **rule-based, documented transformations** on declared inputs, not as unsupervised causal identification from latent structure.

Footnote discipline: Where product copy uses "predictive layer," Knowledge limits it to **24h / 7d** framing **only on surfaces that ship that feature**[^naciro][^dashboard].

[Figure: Data path from Legal Sources Registry through hygiene rules to Naciro stages and NFSI headline — Navy `#1a365d`.]

2.2 LPU architecture and low-latency inference

The **Large Processing Unit (LPU)** architecture entity documents a **specialised inference stack** associated with Naciro, including published **sub-50 ms** inference latency claims and deterministic inference positioning relative to the broader platform[^lpu]. The Knowledge graph intentionally **does not** treat vendor-specific accelerator brands as definitional content; reviewers should consult the **LPU** entity and linked company pages for implementation detail[^lpu].

Scientific readers should separate three layers: (i) **measurement** (inputs), (ii) **algorithmic scoring** (NFSI layers per VVR), and (iii) **inference machinery** (latency-sensitive components for which separate performance evidence may apply).

2.3 Nationfile JSON as the atomic machine-readable profile

Nationfile JSON is the canonical serialisation of a country- or context-level profile: a single logical unit behind charts, badges, NFSI analytics, and exports[^nfjson]. It functions as the **human–machine bridge**: HTML pages and JSON exports share one validated payload definition[^nfjson].

The NFKG repository (`knowledge-data`) publishes a bounded set of **structured entity JSON files**, FAQs, and glossary entries—this manuscript **grounds** terminology in those records rather than in marketing collateral[^hub][^index].

Clarification on scale: Some informal estimates circulate regarding very large entity counts; the **auditable** scope for third-party reviewers is the **published Knowledge index**, bundle timestamps, and export manifests documented in **index.json** and **distribution.json**[^index][^dist].

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3. METHODOLOGY OF HEADLINE INDICES

3.1 NationFiles Stability Index (NFSI)

NFSI is documented as a **0–100** scalar with ordered bands **A–E**[^nfsi]. It is **computed by Naciro** and refreshes on a **15-minute** cycle per Knowledge attributes[^nfsi]. The **VVR** states a **multi-layer** construction (indicator layer, thematic smoothing with inertia, weighted finals including documented **crash mode** handling) and names representative upstream families (e.g. World Bank–class feeds, conflict event corpora, Interpol-class feeds, GDELT-class news risk)[^vvr-md].

NFSI is explicitly **not** a substitute for official national accounts readings where **Purchasing Power Index (PPI)** bundles are concerned, and **Governance & Institutions Index (GGI)** composite semantics are likewise distinct[^ppi][^ggi].

[Figure: NFSI band scale and temporal resolution vs. annual macro series.]

3.2 Governance & Institutions Index (GGI)

GGI is described as a **NationFiles governance bundle** anchored in **Worldwide Governance Indicators–style dimensions**, rolled into one comparable **0–100 composite headline** on dashboards and in PPI/GGI snapshot bundles[^ggi]. Knowledge stresses institutional quality measurement **separately** from GDP-style macro aggregates and denies substitutability for NFSI semantics[^ggi].

3.3 Purchasing Power Index (PPI) and real-economy metrics

PPI summarises a **macro bundle** with **GDP at purchasing power parity per capita** as headline cross-country comparator, together with prices, labour-market indicators, reserves, VAT context, and optional co-bundled NFSI fields where product surfaces combine them[^ppi]. Internal keys (e.g. `gdp_ppp_pc`, `inflation_cpi_pct`) are declared in an application-side metric registry to align UI, snapshots, and exports[^ppi].

Near–daily **foreign-exchange** quotations aggregated for the NationFiles front-end are **distinct series** meant to be read **alongside** slower macro panels[^ppi].

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4. DATA INTEGRITY AND VERIFICATION

4.1 Legal Sources Registry

The **Legal Sources Registry** is a **first-party, citation-forward catalogue** that names **primary sovereign feeds**, official portals, and authoritative government databases underpinning **OSINT-grade** transparency—so readers can verify claims without relying on opaque third-party assertion graphs as primary[^lsr].

Structured disclosure clarifies where citations are expected to land: ministry bulletins, national statistical releases, regulator notices, diplomatic channels, and other **named primaries**, with explicit independence posture relative to vendor-only narratives[^lsr].

4.2 Validation & Verification Report (VVR)

The **VVR** narrates **data lineage** from registry entries through cleansing rules to **Naciro-governed evaluation layers** underpinning NFSI[^vvr]. A declared goal is **traceability density**: completeness of mapping from published inputs through hygiene and scoring stages to headline NFSI—supporting **E-E-A-T** (Experience, Expertise, Authoritativeness, Trust) review[^vvr].

The standalone VVR markdown in this repository elaborates **layered aggregation**, **forecast methodology** (including VAR-based elements with declared history windows where applicable), and **governance** expectations for external auditors[^vvr-md].

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5. PREDICTIVE ANALYTICS AND APPLICATION SURFACES

5.1 Predictive layer (24h / 7d)

Where disclosed, the **predictive layer** provides **24-hour and 7-day** outlook snapshots; Knowledge forbids treating it as a universal headline oracle[^naciro]. Operationally, it supports **briefing horizons**, not long-horizon structural forecasts[^nature][^dashboard].

[Figure: Placement of 24h/7d layer relative to NFSI refresh and annual macro data.]

5.2 Security radar and metamaps

Security Radar and **Meta Maps** constitute **spatial intelligence** layers: thematic map experiences that stack demographic–urban, energy–infrastructure, and publicly documented geopolitical tension lenses so that geographic concentration is visible before tabular detail[^metamaps][^secradar].

Meta Maps inherit a **documentation-bound** ethos: no classified overlays, no sensational shading, and no promise of predictive combat analytics beyond what explicit sources support[^metamaps].

5.3 Country Intelligence Dashboard

The **Country Intelligence Dashboard** aligns NFSI headline stability, curated **sentiment-bearing** news strands, and **security-context radar** in one ISO-facing hub for daily situational awareness[^dashboard]. Outputs include unified briefing surfaces: NFSI score and bands, news overview, security radar context, plus **24h/7d predictive** snapshots where shipped[^dashboard].

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6. ETHICS AND GOVERNANCE

6.1 NationFiles AI Guidelines

The **AI Guidelines** entity ties automated-access expectations, citation of canonical HTTPS pages, and machine-readable export usage under declared licensing posture[^ai][^dist]. Secondary summaries must not override primary abstracts without marking derivation[^pubcorp].

6.2 Transparency and academic citation

The **publications corpus** links long-form methodology artefacts, validation memos, and whitepapers to the same operational stack as live profiles and exports—so citations can move from **footnote → live metric surface → back** without silent definitional drift[^pubcorp].

For DOI-based deposition, publishers should register handles pointing to **immutable version snapshots**; concurrent **built_at_utc** (this document) and **bundle_snapshot_utc** (Knowledge index) should appear in supplementary metadata[^index].

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7. CONCLUSION AND OUTLOOK

NationFiles combines **periodically and—in places—high-frequency refreshed** stability metrics with **declared provenance** and **machine-readable** country profiles. The scientifically interesting claim is not sloganeering about novelty, but **auditability**: whether external reviewers can reconstruct, within disclosed tolerance, the path from registered primaries to headline indices.

Further work for independent scholars includes: (i) empirical comparison against editorially static indices under shock episodes, (ii) sensitivity analysis on documented weights (where published), and (iii) bibliometric tracking of **DOI-registered** NationFiles artefacts against downstream RAG corpora.

REFERENCES (APA 7TH EDITION — ILLUSTRATIVE)

Neawolf Media Group. (2026). *NationFiles Knowledge: Repository index* (Machine-readable dataset). `https://nationfiles.com/en/knowledge/` — companion mirror: `knowledge-data/index.json` (see `bundle_snapshot_utc`).[^index]

Neawolf Media Group. (2026). *NationFiles distribution metadata* (`distribution.json`). Publication ethics, imprint pointers, and citation strings.[^dist]

Neawolf Media Group. (2026). *Validation and Verification Report — NationFiles Stability Index (NFSI)* (VVR). Document ID as cited in repository copy: VVR-2026-03-07.[^vvr-md]

NationFiles Knowledge entities (persistent HTML): individual entries as footnoted below — canonical pattern `https://nationfiles.com/en/knowledge/entity/{id}/` (language prefix may be substituted).

KNOWLEDGE ENTITY FOOTNOTES (CANONICAL EN URLS)

[^naciro]: Naciro Engine

[^nfsi]: NationFiles Stability Index (NFSI)

[^lpu]: LPU architecture

[^nfjson]: Nationfile JSON

[^ggi]: Governance & Institutions Index (GGI)

[^ppi]: Purchasing Power Index (PPI)

[^lsr]: Legal Sources Registry

[^vvr]: Validation & Verification Report (entity)

[^vvr-md]: NFSI Validation and Verification Report (Markdown) — repository twin: `research/nationFiles-Stability-Index--NFSI--Validation-and-Verification-Report.md`

[^dashboard]: Country Intelligence Dashboard

[^metamaps]: Meta Maps

[^secradar]: Security Radar

[^ai]: NationFiles AI Guidelines

[^pubcorp]: Publications corpus

[^hub]: Knowledge hub (NFKG)

[^core]: Core hierarchy

[^index]: Repository: `knowledge-data/index.json`

[^dist]: Repository: `knowledge-data/distribution.json`

[^nature]: Nature / ecology vertical

End of manuscript.

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INDEX COMPARISON & BENCHMARKING

NFSI in the landscape of geopolitical index families

STAND 02.05.2026 · EN

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Index comparison & benchmarking

As of: 02.05.2026

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AT A GLANCE

- The NationFiles Stability Index (NFSI) is recomputed on a documented 15-minute cadence — far more frequent than typical annual indices.
- 190+ countries share the same engine; the public source registry shows which data streams feed the model.
- A short magazine-style briefing comes first; tables and methodology follow for readers who want depth.
- More narrative context in the Knowledge article; technical depth via the Validation & Verification Report (DOI) and the methodology whitepaper.

EVIDENCE & STRATEGIC SOURCES

Persistent identifiers (DOIs), repositories, and mirrors for media, research, and regulators — quick proof of methodology and disclosure.

Zenodo DOI — Validation & Verification Report
Persistent record on Zenodo, ready to cite.
https://doi.org/10.5281/zenodo.19783682

Academia.edu — full VVR text
Academic access point for the verification report.
https://www.academia.edu/166032078/NationFiles_Stability_Index_NFSI_Validation_and_Verification_Report

GitHub — Neawolf Media Group organisation
Code and public repositories.
https://github.com/Neawolf-Media-Group/

GitHub — nationfiles-public-docs
Public documentation and companion artefacts.
https://github.com/Neawolf-Media-Group/nationfiles-public-docs

Hugging Face — Neawolf Media Group
Models, datasets, and AI-facing releases.
https://huggingface.co/Neawolf-Media-Group

Zenodo DOI — LPU architecture
Technical architecture (Large Processing Unit), citable.
https://doi.org/10.5281/zenodo.19774594

nationfiles.com/llms.txt
Machine-readable index of public NationFiles content.
https://nationfiles.com/llms.txt

Legal — Validation & Verification Report (web)
HTML edition on NationFiles, language-aware.
https://nationfiles.com/en/legal/validation-and-verification-report/

Publication — Algorithmic geopolitics (NFSI)
Methodology whitepaper for the geopolitical stack.
https://nationfiles.com/en/publications/algorithmic-geopolitics-methodology/

Legal — Data source registry
Publicly documented connectors and provenance.
https://nationfiles.com/en/legal/sources/

Market context

COUNTRY CONDITIONS IN MINUTES, NOT CALENDAR YEARS: WHY NATIONFILES IS GAINING INTERNATIONAL ATTENTION

Established geopolitical indices deliver essential annual snapshots. NationFiles tackles the adjacent question: what does a situational picture look like when it is engineered to match the rhythm of news? The NationFiles Stability Index (NFSI) runs on a documented 15-minute pipeline, backed by public data streams, a transparently documented connector registry, and an open calculation path — unusual in comparable offerings.

Newsrooms, think tanks, and corporate risk teams operate on two clocks: the publication calendar of classic index houses — and the clock of markets, conflict reporting, and social dynamics. Annual composites excel at long-horizon comparison but hit a wall in real time: the published headline value stays formally frozen until the next release even as conditions on the ground have already shifted.

NationFiles positions the NFSI as an operational instrument: a global stack over documented OSINT and macro connectors, recomputed on a fixed cadence. Instead of an opaque headline that only unlocks with a consultancy mandate, the model emphasises professional traceability — a public source registry, a Validation & Verification Report with DOI, a whitepaper on algorithmic geopolitics, and versionable methodology.

Compared with expensive mandate memos or proprietary terminal feeds, NationFiles offers a different profile: scalable country coverage (more than 190 states under the same engine), consistent entity discipline through the knowledge layer, and API/export paths for teams that need KPIs inside their own scenarios and dashboards — without measurement rules being reinterpreted per client.

Competition for geopolitical data is fragmented: NGO indices deliver academic citation, banks and asset managers deliver market depth, strategy consultancies deliver context — often walled, costly, and hard to replicate. NationFiles combines transparency and speed with a clear product promise: stability- and risk-centric scores grounded in documented rules, running at a cadence aligned with modern news and crisis workflows.

> **What NationFiles bundles today**
>
> - NFSI with documented 15-minute recomputation of publicly shown headline values.
> - Global coverage: more than 190 countries under the same, non-mandate-dependent calculation logic.
> - Wide set of documented data connectors (macro, event, and trend paths) with a public registry at /legal/sources/.
> - Naciro engine: rule-based pipeline (layers 1–3), readable in public methodology — no hidden per-score edits via opaque expert clicks.
> - Validation & Verification Report (Zenodo DOI) and methodological whitepaper as citable technical references.
> - Country pages, regional maps, and security/topic layers — usable for research, briefing, and monitoring.
> - API and export paths (including machine-readable formats) for organisations integrating NFSI and context data into their own tools.

Methodology & market comparison

Knowledge article on methodological benchmarking — The companion article carries the same matrix with additional cross-references.

Composite country scores have become a standard tool in policy research, international reporting, and risk governance. Depending on their mandate, established indices capture state fragility, violent conflict, regime quality, perceived corruption, or governance aggregates — each with defined input logic, field windows, and publication discipline.

The NationFiles Stability Index (NFSI) addresses the same translation problem — country stability and risk — but commits to a different yardstick: a documented, rule-based computation pipeline with high cadence and a public connector registry. It does not substitute primary scholarship on any third-party index cited here; it transparently situates the NationFiles system within the existing field of geopolitical measurement instruments.

The text follows a comparative reporting register: no vendor league tables, no procurement guidance, and no legal appraisal of third-party products. Primary methodology publications by each publisher remain authoritative.

LATENCY AND PUBLICATION RHYTHM

- NationFiles / NFSI: documented recomputation on a 15-minute cycle; published headline values therefore refresh well below the one-hour mark.
- Many classic country and governance indices: typically annual releases; as a didactic order-of-magnitude comparison, symbolically up to about 8,760 hours between comparable full snapshots.

Exact field and release timing must be taken from each provider's primary documentation.

GLOBAL BENCHMARK MATRIX (FUNCTIONAL)

Mapping by typical sponsor, data basis, order of magnitude of updates, and thematic focus. Actor labels follow international usage.

PROVIDER / LINE	SPONSOR (TYPICAL)	DATA BASIS (SIMPLIFIED)	UPDATE (ORDER OF MAGNITUDE)	PRIMARY FOCUS
NATIONFILES (NFSI)	Neawolf Media Group	Public OSINT feeds, documented connector registry	High-frequency (documented 15-minute recomputation)	Stability and country risk
FRAGILE STATES INDEX (FSI)	Fund for Peace	Qualitative and quantitative country profiles	Annual	State fragility
GLOBAL PEACE INDEX (GPI)	Vision of Humanity	Surveys, expert panels, secondary sources	Annual	Peacefulness and violent conflict
DEMOCRACY INDEX (EIU)	Economist Intelligence Unit	Survey and indicator aggregation	Annual	Regime and institutional quality

PROVIDER / LINE	SPONSOR (TYPICAL)	DATA BASIS (SIMPLIFIED)	UPDATE (ORDER OF MAGNITUDE)	PRIMARY FOCUS
CPI (TRANSPARENCY INTERNATIONAL)	Transparency International	Expert panels, secondary data	Annual (typical)	Perceived corruption
WORLD GOVERNANCE INDICATORS (WGI)	World Bank	Aggregation of surveys and sources	Annual or data-dependent	Governance and institutions
STRATEGIC ADVISORY (E.G. RANE)	Commercial	Expert networks, mandate-led research	Subscription- or case-dependent	Strategic assessment
CONTROL RISKS ET AL.	Commercial	On-the-ground networks, consultant models	Mandate-dependent	Corporate and site security
COUNTRY RISK RESEARCH (E.G. IHS / S&P)	Commercial	Macro and market research	Product-line dependent	Market and credit context
BLACKROCK GPR / SIMILAR NEWS INDICES	Institutional vendor	Text mining of news flows	Typically monthly to high-frequency	Market-relevant risk signals

Functional classification only — no implicit quality or ranking claim.

THREE FUNCTIONAL CLUSTERS

CLUSTER	LOGIC	TYPICAL EXAMPLES
A — Academic and NGO indices	Structural long-horizon diagnostics, wide citation reach	FSI, GPI, EIU Democracy Index, CPI, WGI
B — Commercial strategy advisory	Discretionary, mandate-bound intelligence	RANE, Control Risks, IHS / S&P Global (product lines)
C — Market-linked quantitative trackers	Market- and text-driven risk indicators	BlackRock GPR, Bloomberg terminal signal paths

SECTOR ANALYSIS: TYPICAL PROFILES

Qualitative positioning relative to professional situational-awareness markets — not an absolute valuation in currency terms.

A — Academic and civil-society indices

State resilience, conflict and peace, democratic quality, perceived corruption, governance signals.

CRITERION	TYPICAL PROFILE
Latency	Very high: yearly publication windows; symbolically up to about 8,760 hours between comparable full releases.
Cost	Low to medium: outputs often free or academically accessible; specialised feeds sometimes fee-based.
Transparency	Medium to high: methodology reports usually public; raw micro-data sometimes restricted for third parties.

Typical strengths

- High uptake in research and policy routines.
- Long time series and cohort-style country comparisons.
- Definitional frames documented in primary methodologies.

Typical limits

- Limited intra-year operational refresh.
- Headline values remain static until the next full release.
- Often too slow for tactical crisis guardrails.

B — Commercial strategy and risk analysis

Mandate-bound assessment, on-ground signals, corporate and site-protection logic.

CRITERION	TYPICAL PROFILE
Latency	Medium: subscription- or event-driven, without one globally uniform grid.
Cost	High: advisory and licensing models.
Transparency	Low to medium: proprietary models; details often confidential.

Typical strengths

- Deep human expertise and context-sensitive narratives.
- Action-adjacent briefings for defined mandates.
- Rapid spot assessments with close client coupling.

Typical limits

- Hard to replicate for outside auditors.
- Equal treatment of all jurisdictions under one identical engine is rarely the objective.
- Cross-client comparability is constrained.

C — Financial-market and quantitative risk trackers

Market reactions, text and news signals, terminal and feed economics.

CRITERION	TYPICAL PROFILE
Latency	Low to medium: monthly composites or high-frequency data paths.
Cost	High to very high: institutional licences.
Transparency	Low to medium: trade secrecy versus partial whitepapers.

Typical strengths

- Higher frequency than pure annual indices.
- Coupling with market and trading data.
- Plugs into portfolio and treasury workflows.

Typical limits

- Emphasis often on financial markets; institutional full pictures vary by product.
- Definitions of “risk” differ across product lines.
- Societal and hard-security layers are not always equally deep.

NFSI: YARDSTICK COMPARISON

Reference: typical annual indices with a symbolic ~8,760-hour gap between comparable headline releases.

DIMENSION	NATIONFILES / NFSI	TYPICAL ANNUAL INDICES (~8,760 H WINDOW)
FRESHNESS	Documented 15-minute recomputation of published headline values.	Full refresh typically yearly; shocks within the survey year may not yet appear in the headline value.
AUDITABILITY	Rule-led paths via the public registry, VVR, and whitepaper.	Methods often public; raw engine or micro-data partially closed to outsiders.
SYSTEM CONSISTENCY	Entity and namespace discipline reduces semantic drift of core terms.	Project-specific holdings; definitions may shift between editions.

Reading guide: a distinct technical target — high freshness, auditable computation paths, disciplined terminology — rather than moral superiority.

METHODOLOGICAL BOUNDARIES

- The NFSI is not an opaque one-click oracle but a rule-based aggregate with a documented pipeline (Naciro engine).
- Score changes should trace to inspectable rules and evidenced inputs — not to opaque expert nudging in production.
- Explicit country aggregation distinguishes the system from bare news-sentiment feeds without a population frame.
- The Governance Institutions Index can cover institutional context that sentiment-only trackers underspecify.
- Where documented in the VVR and whitepaper, reproducible audit trails and robustness evidence contrast with the retrospective validation rhythms of many annual composites.

CITATIONS AND PRIMARY SOURCES

The NFSI pipeline points to the Validation & Verification Report (Zenodo DOI), the methodological whitepaper, and the public source registry. This page does not replace primary literature for any third-party benchmark cited.

- DOI 10.5281/zenodo.19783682 — Validation & Verification Report
- Publication: algorithmic geopolitics and NFSI methodology
- Public data-source registry (Legal / Sources)

- Legal: Validation and Verification Report

COMPACT COMPARISON

CRITERION	NATIONFILES / NFSI	TYPICAL ANNUAL INDICES
LATENCY	15-minute cadence (documented)	typically yearly release
DATA PATH	OSINT / connector aggregation with registry	often expert and survey cycles
TRANSPARENCY	public registry, VVR, DOI	methods often public; raw engine partially closed

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